

Solar Roadways

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Executive Summary

This report evaluates the strategic viability of solar roadway technology as a multifunctional infrastructure solution that integrates renewable energy generation with transportation surfaces. While the concept offers a compelling response to growing climate and energy demands by utilizing existing paved surfaces, our analysis finds that solar roadways are not yet competitive with traditional alternatives such as rooftop or utility-scale solar due to high costs, lower efficiency, durability concerns, and limited long-term performance data.

The most attractive near-term opportunity lies in niche, low-traffic applications, including sidewalks, bike paths, campuses, and parking areas, where the technology's multifunctional benefits such as embedded lighting, heating, and sensors provide added value beyond energy generation. In these environments, solar roadways can be positioned as premium, sustainability-driven infrastructure rather than a direct replacement for conventional roads or solar systems. Market entry should focus on institutional and private-sector customers with strong sustainability mandates and the flexibility to pilot emerging technologies.

From a competitive standpoint, the industry is shaped by established infrastructure firms and alternative smart roadway technologies. Solar Roadways currently differentiates through product innovation but lacks operational scale and institutional partnerships. Partnering with an experienced infrastructure company, particularly Colas, emerges as the most viable commercialization pathway, given its expertise in pavement engineering, existing government relationships, and prior experience with solar pavement systems.

Financial analysis and market projections suggest gradual adoption, with meaningful market share achievable only through phased deployment and demonstrated performance improvements. Key risks include high upfront capital requirements, regulatory barriers, and uncertainty around lifecycle costs and maintenance complexity.

Strategically, solar roadways should be pursued as a long-term innovation platform rather than an immediate large-scale solution. Success depends on focusing on durability, modular design, and pilot validation, while leveraging partnerships and sustainability-driven funding. Under this approach, solar roadways can evolve from experimental concept to viable complementary infrastructure over time.

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Macroeconomic/Societal Environment

Crisis identification

Solar roadways are driven by the growing crisis of climate change and rising energy demand, along with the need to use land and infrastructure more efficiently. As electricity consumption increases due to population growth, electric vehicles, and technology use, governments are under pressure to expand renewable energy sources without consuming additional land. At the same time, roads cover millions of miles of already-developed surface area and require constant public funding, yet they serve only a single function. Solar roadways present an opportunity to turn this existing infrastructure into a source of clean, renewable energy, helping address sustainability goals while avoiding the land-use challenges faced by traditional solar farms (Hyder, 2025).

However, solar roadways also emerge from a crisis of infrastructure inefficiency and resilience. Many roads in the United States are aging, expensive to maintain, and vulnerable to extreme weather events such as heat waves, snowstorms, and flooding. Solar roadway concepts propose added benefits like built-in heating to reduce ice, embedded sensors for monitoring road conditions, and decentralized energy generation that could support nearby infrastructure during power outages. While the technology faces significant challenges related to cost, durability, and efficiency, the increasing strain on transportation and energy systems creates a compelling reason to explore more innovative and multifunctional infrastructure solutions.

Identification of innovation enablers or inhibitors

Society continues to demand more forms of green energy, positioning solar roadways and photovoltaic pavement as a conceptually attractive extension of existing renewable technologies. This demand is reinforced by the steady decline in photovoltaic material costs over the past several decades, with global average solar PV module prices falling dramatically year over year, reducing one of the most historically significant barriers to solar adoption (Ritchie, Rosado, & Roser, 2023). As costs decrease, the feasibility of experimenting with nontraditional PV applications such as road-integrated systems becomes more probable, particularly when the technology offers dual functionality by enabling energy generation without requiring additional land use. Teh, Adebajo, and Kong identify dual functionality as a key motivational factor, noting that solar paver technology “provides an excellent avenue for dual usage of infrastructure,” allowing otherwise underutilized, sun-exposed surfaces to be repurposed for renewable energy generation (Teh, Adebajo, & Kong, 2021). In their focus group and interview study, several participants emphasized that solar pavers enable value creation from exposed surfaces already dedicated to transportation or development projects, with one participant stating, “This kind of product for building industry is a new innovation. It will help to

generate and use up the expose surface” (Focus group session 4) (Teh, Adebajo, & Kong, 2021). In addition, policy and institutional alignment is a key precondition for the development of solar roadways, since these systems depend on public funding, regulatory approval, and government-led infrastructure projects. Teh, Adebajo, and Kong show that financial incentives strongly influence support for solar technologies, with many participants citing government programs such as feed-in tariffs and tax incentives as reasons for adopting solar power (Teh, Adebajo, & Kong, 2021). Favorable climates also matter, as regions with high year-round solar exposure are more likely to justify public investment by demonstrating consistent energy generation (Teh, Adebajo, & Kong, 2021). Together, supportive policies, financial incentives, and suitable environmental conditions create an environment in which solar roadways can be tested and evaluated, even though these factors alone are not enough to ensure widespread adoption.

Several significant inhibitors limit the practicality of solar roadways, particularly when viewed through the lens of construction and long-term infrastructure management. A major barrier is reluctance from the government, as transportation and municipal agencies often lack the willingness to maintain unconventional infrastructure, leading to rejection in favor of proven solutions such as solar panels that are just lined along roadways. High upfront costs further discourage adoption, since photovoltaic pavement must be strong enough to withstand traffic loads, weather exposure, and waterproofing requirements, all of which substantially increase construction expenses (Teh, Adebajo, & Kong, 2021). Environmental conditions potentially pose the most serious challenge, especially in regions with high humidity and rainfall, where water seepage, material degradation, and sealing failures raise concerns about durability and safety. As one focus group put it, “And comparing to that existing solution, roof top solar don't have to consider all these, relatively the maintenance is lower than pavement because it doesn't undergo all the load” (Teh, Adebajo, & Kong, 2021). In addition, the complexity of installation and maintenance makes repairs difficult, as failures may require replacing entire sections rather than individual components. Public skepticism and concerns about low efficiency and questionable practicality further inhibit adoption, particularly when compared to established alternatives such as rooftop solar, which offer higher efficiency, lower maintenance, and better lifecycle economics. Together, institutional resistance, construction complexity, environmental exposure, and weak comparative advantage create substantial barriers that limit the feasibility of solar roadways as a scalable infrastructure solution.

Timing

Although solar roadways represent an innovative approach to renewable energy, the world is not yet ready for their widespread adoption due to economic, durability, and safety limitations that must be resolved before large-scale implementation is possible. These roadways are promising, projected to produce “three times our annual energy consumption”, light up dark roads, and charge electric vehicles (Marsh, 2023). However, the first implementation of solar roadways in Idaho failed miserably from a fire in its electrical system. In addition, another solar roadway installation was created in Normandy, France in 2016. The installation proved to be unsuccessful in withstanding

traffic and generated much less energy than expected, resulting in a termination in 2019 (Colagrossi, 2019).

The current issues with solar roads include high costs, durability, less energy production due to shading and flatness, noise concerns, and excess heat (Shekhar, et al., 2018). Before solar roadways can be adopted on a wider scale, these problems must be addressed and fixed, with long-term testing conducted by third-party research organizations to demonstrate durability under heavy traffic and varying climates. Additionally, costs must be reduced to compete with traditional solar panels, which currently generate more energy at lower prices. Government transportation agencies would also need to establish safety, maintenance, and cybersecurity standards before approving solar road installations.

One solar bike path installed in the Netherlands was successful, generating about 3,000 kilowatt-hours (kWh) of electricity (Shekhar, et al., 2018). This example suggests that solar roadways may work best on low-intensity surfaces, such as bike lanes or slow-speed roads, rather than major highways. Overall, while solar roadways are not yet practical for widespread use, using them in low-traffic environments could allow the technology to improve and be tested further before broader adoption becomes realistic.

Strategic implications

The analysis above shows that while photovoltaic pavement is an interesting and creative idea, it is not yet well aligned with the main factors that typically drive large-scale adoption. As shown in Figure 1, the factors with the highest relative importance—crisis intensity and existing alternatives—do not currently favor photovoltaic pavement. Although climate change and increasing energy demand are serious concerns, photovoltaic pavement does not offer a clear advantage in addressing these issues when compared to more established and cost-effective options such as rooftop or utility-scale solar. This suggests that, at present, photovoltaic pavement does not respond to a unique or urgent crisis in a way that would justify widespread use.

Figure 1 also shows that existing alternatives present a significant challenge. These alternatives are highly important and perform better in terms of cost, durability, efficiency, and ease of maintenance. In comparison, photovoltaic pavement has a relatively low advantage, which makes it difficult to justify as a direct replacement for conventional solar technologies, especially on high-traffic roads or major transportation routes.

On the other hand, photovoltaic pavement performs somewhat better in areas such as societal readiness and regulatory readiness. These factors show moderate to high relative advantage, suggesting that the public and policymakers may be open to new, sustainability-focused infrastructure ideas. However, this general openness is not enough on its own to overcome the technical and economic challenges identified earlier. Other enablers, such as declining photovoltaic costs and interest in dual-use infrastructure, help support development, but their lower importance indicates that they play a supporting role rather than driving adoption on their own.

Overall, the matrix suggests that the current vision for photovoltaic pavement needs to be adjusted to better fit current conditions. Instead of aiming for immediate large-scale deployment, the technology is better suited for smaller, more controlled applications where its benefits are more meaningful. Locations such as bike paths, pedestrian areas, parking lots, and campus roads offer lower traffic speeds and loads, which reduces safety concerns and allows for real-world testing without high risk.

To improve the chances of long-term success, innovators should focus on designing modular panels that are easier to replace and maintain, running pilot projects in regions with strong solar potential and limited land availability, and working closely with public agencies to evaluate long-term performance. Until photovoltaic pavement can match existing solar technologies in terms of cost and reliability, it should be viewed as a complementary solution rather than a primary source of renewable energy. Under these conditions, continued development makes sense, but widespread adoption should wait until key technical and economic challenges are resolved.

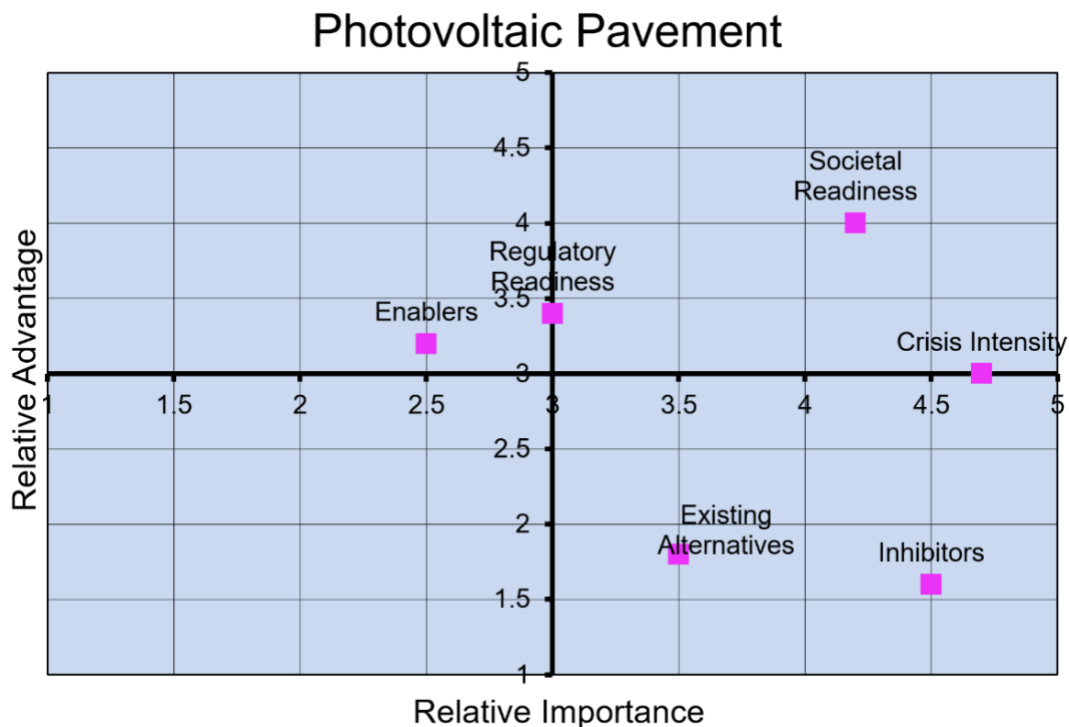


Figure 1: Importance-Advantage Matrix for Macro-Societal Innovation Success Factors.

Market/Demand Environment

Ideal market for your technology

The ideal market for solar roadways is the public infrastructure and transportation sector, with an initial focus on private developers and property owners, particularly those responsible for parking lots, sidewalks, and pedestrian areas in commercial spaces such as shopping centers, corporate campuses, and mixed-use developments. While everyday drivers and pedestrians are the end users of roads, the direct customers for solar roadway technology would most likely be public agencies or private infrastructure contractors responsible for road construction, maintenance, and energy planning. These organizations already manage large transportation budgets and are increasingly required to meet sustainability, resilience, and safety goals. By targeting areas where roads are already scheduled for repair or replacement, solar roadways could be integrated into existing infrastructure projects rather than requiring entirely new development.

Within this broader market, the most promising early applications are low-traffic, high-value environments such as parking lots, sidewalks, bike paths, campuses, airports, and military bases. These settings reduce wear and durability concerns while allowing users to benefit from on-site energy generation, smart sensors, and safety features like ice prevention and lighting. Over time, as materials improve and costs decrease, the technology could expand into higher-traffic roadways and highways. Although solar roadway developers may license the technology to construction firms or materials manufacturers, the ultimate end market remains public and institutional infrastructure seeking multifunctional, sustainable solutions that combine transportation, energy generation, and resilience in a single system.

Candidate markets

Parking lots and garages, elevated highways and roadways, and college campus sidewalks are three potential markets that have many of the ideal characteristics described above for Solar Roadways. Each of these markets represents a large, paved area that is increasingly under pressure to find ways to integrate renewable energy solutions and sustainable technologies, but they are vastly different in terms of traffic, regulatory issues, and overall feasibility of early market entry for Solar Roadways. The most attractive initial entry market for Solar Roadways is road and highway construction contractors, specifically those responsible for pedestrian-oriented infrastructure such as sidewalks, bike trails, shared-use paths, and private roadways. This market segment represents a broad range of construction activities, including residential and non-residential construction, and is best represented by the Highway, Street, and Bridge Construction industry. According to IBISWorld, road and highway construction activity increased through the end of 2025 due to strong investment in transportation infrastructure, with industry revenue expanding at a compound annual growth rate of 3.0% over the past five years and reaching an estimated \$193.4 billion in 2025 (Pigott, 2025). This growth has been supported by state and local transportation budgets as well as federal programs such as the Highway Trust Fund and the Infrastructure Investment

and Jobs Act, which allocated approximately \$350 billion toward federal highway programs through (Pigott, 2025). IBISWorld projects industry revenue to continue expanding at a CAGR of 1.1% to, driven by increased travel demand and recovery in private nonresidential construction, including airport runways, private roads, and institutional infrastructure (Pigott, 2025). This market also includes college campus sidewalks and pedestrian infrastructure, which are optimal entry points for Solar Roadways due to lower traffic rates, higher solar availability, and strong institutional motivations to pursue sustainability initiatives. Unlike highways or parking garages, college campuses permit construction contractors to test new technologies in optimal settings while meeting client demands for carbon reduction and renewable energy solutions. Moreover, the large scope of the industry, with over 44,000 companies in the United States, provides opportunities for early adoption by large civil engineering firms looking to differentiate themselves through sustainable construction practices (Hoovers, n.d.). Overall, the strength of infrastructure spending, the continued growth of the industry, and the increasing demand for sustainable pedestrian infrastructure make road and highway construction contractors a compelling initial market opportunity for the adoption and scaling of Solar Roadways technology.

Market size

The market size that was selected for Solar Roadways was based upon the U.S. highway, street, and bridge construction market, which represents the market of contractors that build and resurface roadways, sidewalks, bike paths, and other paved transportation infrastructure. As previously mentioned, this industry earns a revenue of about \$193.4 billion within the 2025 market year (Pigott, 2025). However, due to the early stage of Solar Roadways and the technical constraints of serving a high-traffic vehicular market, the realistically served market size represents a small fraction of this figure. With respect to the initial market entry, the served available market segment was further segmented to include pedestrian infrastructure such as sidewalks, bike paths, shared use paths, and private roadways designed and built by institutional and commercial contractors. Pedestrian infrastructure such as sidewalks represents only a small percentage of overall highway and road construction activity that is estimated to be approximately 1% of the total industry and thus requires a conservative approach to initial market sizing (Pigott, 2025). As indicated in Figure 2 below, the initial served market share is estimated to be below 1% of the overall industry. In the future, the served market segment is expected to grow as the technology advances and expands from merely being used in sidewalks to other low load transportation surfaces. As the technology develops and proves its capability, the served market segment is expected to represent a larger percentage of the overall industry in the future. Despite the fact that the initial served market segment represents only below 1% of the overall industry, the size of the overall industry is large. Industry growth is also expected to continue beyond the initial entry period, with IBISWorld projecting highway and street construction revenue to increase at a compound annual growth rate of 1.1% through 2030, reaching approximately \$204.7 billion, driven by increased travel demand and recovery in private nonresidential construction, including airports, campuses, and private transportation infrastructure (Pigott, 2025). While Solar Roadways is unlikely to impact a large portion of the total market in its early years, the combination of a large underlying construction

market, sustained infrastructure funding, and growing demand for sustainable materials supports the feasibility of entering the market at a limited scale and expanding adoption over time as production capacity and performance validation improve.

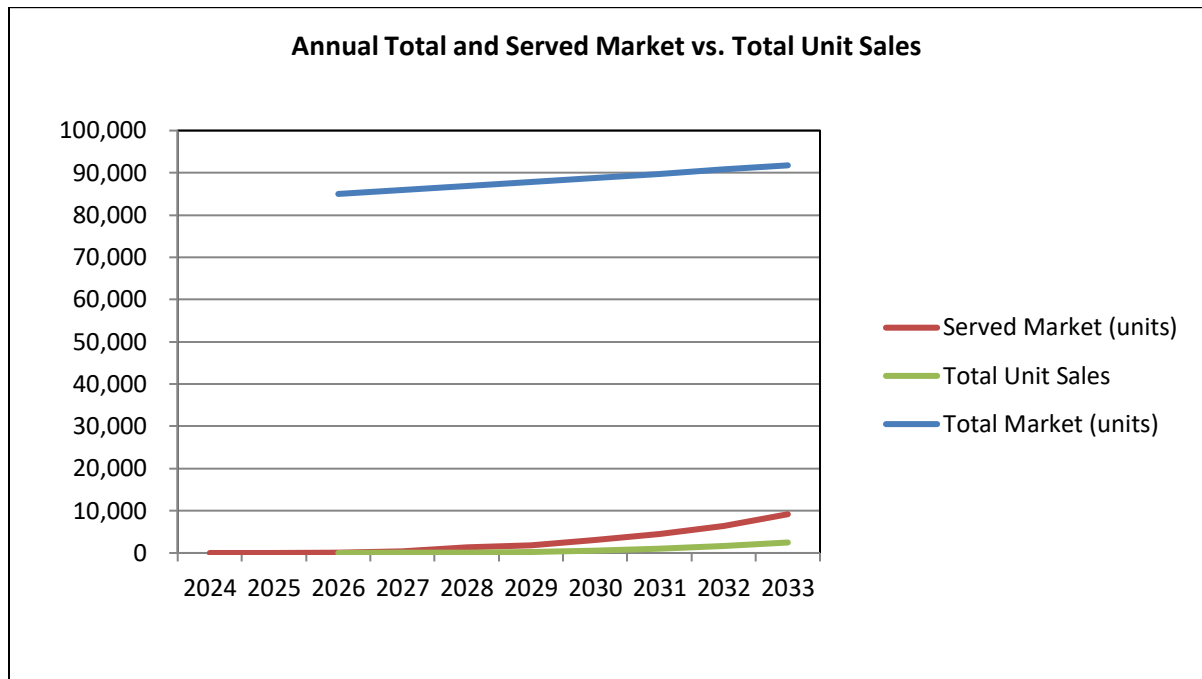


Figure 2: Importance-Advantage Matrix for Macro-Societal Innovation Success Factors.

Market's Needs/Expectations from the Technology

From the customer's perspective, solar roads are only practical if they first function as reliable pavement. Since the main market of solar roads would be cities and large property owners who are held responsible for the safety of their inhabitants, this technology must be thoroughly tested. Any solar roadway must provide dependable traction in rain and snow, withstand its intended traffic loads, and allow for predictable maintenance without frequent closures. If the surface fails as pavement or creates new maintenance issues, customers will not care how innovative the technology is.

An important expectation for solar roads is their efficient use of space. Many potential customers already control large areas of pavement such as parking lots, sidewalks, bike paths, and private roads, but do not have sufficient roof area or nearby land for traditional solar installations. Solar roads appeal because they generate electricity on surfaces that already exist and are already budgeted for replacement over time. Pilot projects have shown that solar pavements can produce meaningful amounts of energy, roughly 70 kilowatt-hours per square meter per year, which can be used locally for lighting, signage, sensors, or nearby electric vehicle chargers (Bazley, 2015). While this output is lower than conventional solar panels, customers may still value the ability to generate power directly where it is needed without acquiring new land.

Another important expectation is the potential to reduce winter maintenance and safety risks. Snow and ice management are costly and labor-intensive, especially in

colder regions where it also creates liability concerns. Solar road systems that incorporate heating functionality could reduce plowing, salt use, and time of workers if they perform consistently. This expectation is grounded in the fact that customers already pay premiums for heated driveways and walkways, which are commonly installed at costs between roughly \$7-\$16 per square foot for the heating system alone (Billen, 2026). For customers, the appeal of solar roads is therefore not just sustainability, but the possibility of combining energy generation with safer, lower-maintenance infrastructure.

Pricing – customer willingness to pay

Customer willingness to pay for solar roads is best understood by looking at what they already spend on infrastructure. Public agencies and large property owners routinely budget for pavement resurfacing and rehabilitation, which can cost hundreds of thousands of dollars per lane-mile depending on location and scope (Federal Highway Administration, 2020). This establishes a realistic baseline for what customers consider acceptable spending on high-quality pavement. The cost of solar roads is evaluated as an upgrade to an expense that customers already expect to incur.

Simultaneously, customers are very aware of the cost of conventional solar energy. Traditional photovoltaic systems are far more efficient and significantly cheaper on a per-watt basis than solar pavement technologies (National Renewable Energy Lab, 2024). This comparison limits how much customers are willing to pay for the energy generation aspect of solar roads alone. When energy output is translated into dollars, solar pavement currently produces well under \$1 per square foot per year in electricity value at average U.S. electricity prices (U.S. Energy Information Administration, 2025). As a result, customers expect solar roads to deliver additional value beyond electricity generation since there is a higher upfront cost.

The most comparable alternative to solar roads is the use of traditional pavement alongside conventional photovoltaic solar systems. While standard paving and solar installations are relatively cost-effective and well established, early solar road projects have demonstrated substantially higher costs. For example, France's Wattway installation reportedly required an investment of approximately €5 million to construct about one kilometer of roadway, equating to nearly €1,800 per square meter of surface area (Anthony, 2016). Although these costs reflect pilot-scale development rather than long-term commercial pricing, they highlight the need for solar roads to be applied selectively. In practice, solar roads make the most sense as a premium option in controlled, low-speed spaces like parking lots, campuses, and pedestrian areas. In these settings, generating electricity directly on site while serving multiple infrastructure purposes can reasonably justify a higher cost than standard pavement alone.

Triggers and barriers to adopting this technology

The adoption of solar roadway technology in public and institutional infrastructure markets depends on several innovation success factors (ISFs) that act as either triggers or barriers. These factors are defined below in the context of solar roadways and their target market of contractors, campuses, and pedestrian infrastructure.

Relative advantage refers to how much better solar roadways perform compared to traditional pavement combined with separate solar panel installations. For cities, campuses, and infrastructure managers, this includes lifecycle cost savings, added functionality, safety improvements, and the ability to generate energy using already-developed paved surfaces. Solar roadways can produce electricity on land that cannot support rooftop or ground-mounted solar, which is attractive in dense or land-constrained environments (Bazley, 2015). In addition, integrated heating systems could reduce snow and ice removal costs, which are already addressed in other infrastructure through expensive heated pavement systems (Billen, 2026). However, conventional photovoltaic systems remain significantly more efficient and less expensive on a per-watt basis than embedded solar pavement (National Renewable Energy Lab, 2024). Because customers can install traditional pavement and rooftop solar at lower cost and higher performance, the relative advantage of solar roadways is only strong when multiple benefits—such as lighting, heating, and sensors—are valued together rather than energy production alone (U.S. Energy Information Administration, 2025).

Strategic implication: Solar roadways should be marketed as multi-functional smart infrastructure, not simply as an alternative solar technology.

Compatibility describes how well solar roadways fit with existing construction practices, contractor capabilities, procurement systems, and maintenance routines. Road construction today relies on standardized asphalt and concrete methods, and public agencies are accustomed to materials with decades of performance data (Federal Highway Administration, 2020). Solar roadways introduce unfamiliar components such as embedded electronics and energy systems, which may require new training and maintenance procedures. This lack of compatibility with established road-building norms can slow adoption.

Strategic implication: Designing modular systems that resemble existing paving processes and providing contractor training can improve compatibility and reduce resistance.

Trialability is the degree to which the technology can be tested on a small scale before full adoption. Solar roadways are well suited for pilot installations in limited areas such as sidewalks, plazas, bike paths, or parking stalls, which lowers risk compared to deploying them on highways. These environments allow institutions like universities and corporate campuses to test performance while supporting sustainability goals. However, even small pilot installations may be more expensive than conventional pavement, which can limit experimentation (Anthony, 2016).

Strategic implication: The company should emphasize small demonstration projects and collect performance data to build credibility and reduce perceived risk.

Service intensity refers to the level of technical support and maintenance required after installation. Solar roadways combine civil infrastructure with electrical and digital systems, potentially increasing the need for specialized service compared to traditional pavement. Public works departments may be concerned about relying on proprietary vendors for long-term maintenance. Modular, replaceable panels and remote monitoring could reduce this burden, but the perceived service complexity remains a barrier.

Strategic implication: A strong service model, including warranties and local maintenance training, will be necessary to reassure buyers.

ISFs

Solar roadway adoption depends on how well the technology fits within public and institutional funding structures. Infrastructure projects are typically budgeted around conventional resurfacing costs, making higher upfront costs difficult to justify. Early solar road pilots, such as France's Wattway project, demonstrated very high installation costs relative to standard pavement (Anthony, 2016). However, projects that combine transportation, renewable energy, and sustainability goals may qualify for multiple funding streams, including environmental or smart-city grants.

Strategic implication: Position solar roadways as eligible for sustainability and resilience funding, not solely transportation budgets.

Public agencies are highly sensitive to safety and liability. If solar roadways are perceived as slippery, fragile, or prone to malfunction, decision-makers will avoid them. Conversely, if proven to improve visibility and reduce ice buildup, they could lower accident risk and liability exposure. Because infrastructure agencies typically favor proven materials and standards, the lack of long-term performance data is a major barrier (Federal Highway Administration, 2020).

Strategic implication: Extensive testing, certification, and third-party validation are essential before large-scale adoption.

Road infrastructure must meet strict design and safety standards at local, state, and federal levels. Because solar pavement is a new category, it may not fit easily within existing codes and standards, leading to approval delays. Pedestrian and low-speed environments face fewer regulatory hurdles than highways, making them more practical early markets.

Strategic implication: Focus early deployment on sidewalks, campuses, and pedestrian zones while working with regulators to develop standards.

Strategic implications

From the analysis above, it should be clear that the opportunity with solar roadways does not lie with traditional roads but with sidewalks, campuses, plazas, bike paths, and private or institutional developments as a multifunctional smart infrastructure solution in

a controlled and low-risk environment. Here, the limitations and risks associated with solar roadways will be less constraining, while their benefits will be more relevant and appealing. In such environments, decision-makers will be more willing to fund projects, allow experimentation and risk-taking, and invest in non-traditional value propositions related to climate change, innovation branding, and smart cities. Most importantly, these will be opportunities to collect improvement data and learn, which will be critical in establishing credibility and expanding to more challenging applications in the future.

At the same time, the most significant barriers are also those associated with the structural nature of infrastructure markets. These factors include high up-front costs compared to conventional pavement, limited long-term performance history, and maintenance needs, all of which conflict with the traditional infrastructure planning and procurement process. Regulatory issues and safety concerns also contribute to the slower adoption rate, especially for vehicular infrastructure, which has the most significant liability risks. Even if technical performance is acceptable, these barriers represent significant hurdles. Therefore, the major challenge is not technical feasibility, but making the technology fit with existing funding models, regulatory systems, and risk tolerance. To successfully overcome these barriers will be the ultimate test of solar roadways as more than a niche technology.

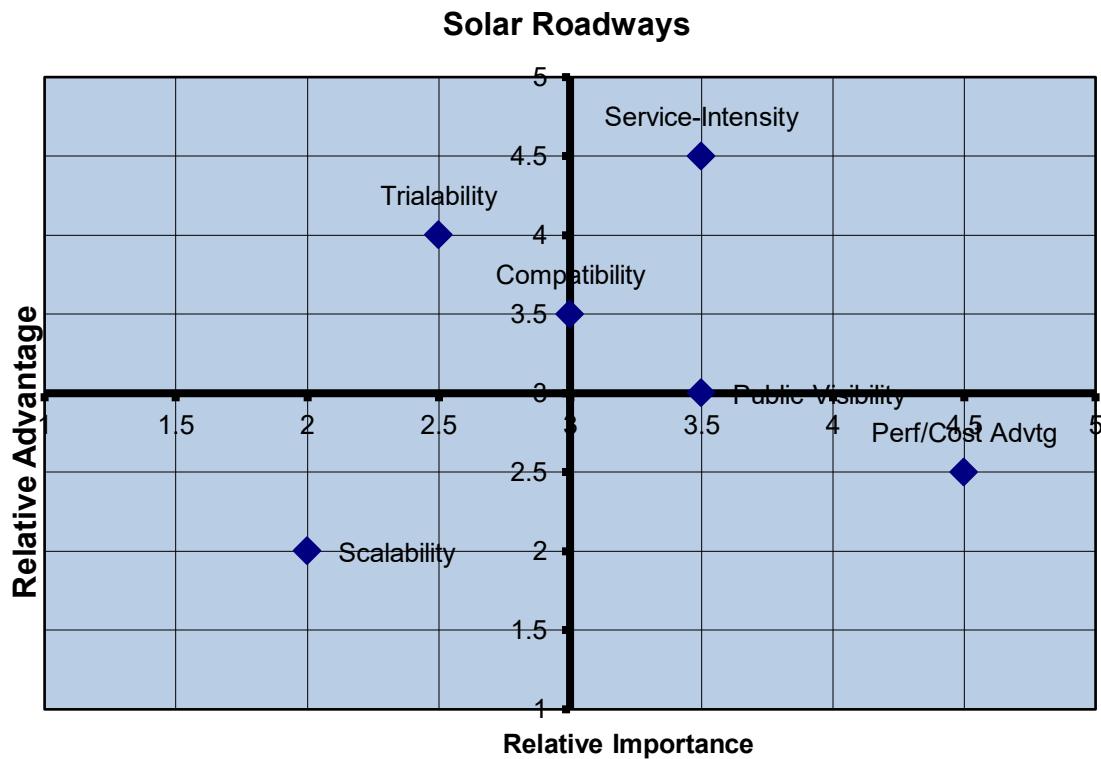


Figure 3: Solar Roadways Importance-Advantage Matrix

Technological Environment

History of the technology

For most of the twentieth century, roadway infrastructure and energy infrastructure developed along separate technological paths. Roads were engineered primarily for durability, load-bearing capacity, and safety, relying on asphalt and concrete pavements that required periodic resurfacing and maintenance. Enhancements to roadway functionality, such as lane visibility or winter safety, were typically addressed through external additions including painted markings, reflective studs, roadside lighting, and in colder climates, electrically heated pavement systems. These snow-melting systems were effective, but energy intensive and rarely implemented at scale due to high operating costs and infrastructure complexity (Pomerantz, Pon, Akbari, & Chang, 2000).

At the same time, electricity generation evolved independently, with solar photovoltaics gaining traction during the late twentieth century as costs declined and efficiency improved. Solar deployment focused on rooftops, parking canopies, and utility-scale arrays installed on undeveloped land, while transportation surfaces remained largely untapped as energy-producing assets. While intelligent transportation systems emerged in the 1990s and 2000s, incorporating sensors, cameras, and electronic signage, these technologies were layered onto existing roadways rather than integrated into the pavement itself (Barbaresso, et al., 2014). The idea of solar roadways arose from the recognition that transportation infrastructure occupies vast areas of sun-exposed land that has the potential to be used for energy production. Early conceptual work explored whether photovoltaic panels could be engineered to withstand vehicular loads while maintaining sufficient traction and optical transmission. Research sponsored by the U.S. Department of Transportation examined the feasibility of replacing traditional pavement with modular solar panels that could simultaneously support traffic and generate electricity (U.S. Department of Transportation, 2022). This approach represented a shift away from single-function infrastructure toward multifunctional systems that combine transportation, energy, and communication roles.

Momentum around solar roadways increased in the early 2010s, driven by federal research funding and increased public interest in renewable energy innovation. The Solar Roadways project gained significant attention after receiving grant support and later through public crowdfunding, which enabled the construction of multiple prototype generations and small-scale pilot installations (U.S. Small Business Administration, 2017). These pilots provided valuable data on durability, power output, and environmental exposure under real-world conditions.

However, broader experimentation with solar pavement technologies internationally highlighted persistent challenges. Flat, ground-level photovoltaic

installations were shown to produce less energy per unit area than conventional tilted arrays and were more susceptible to shading and mechanical degradation (Pasley, 2019). As a result, solar roadway technology has not yet replaced pavement systems and remains primarily in the pilot phase. Current development efforts emphasize design refinement, cost reduction, and smaller-scale applications such as sidewalks, bike lanes, and parking areas. While Solar Roadways has identified its next-generation panel as a commercial product, no definitive large-scale launch timeline has been publicly confirmed, reflecting the cautious transition from experimental concept to deployable infrastructure (Solar Roadways, SR4 is Here, 2019).

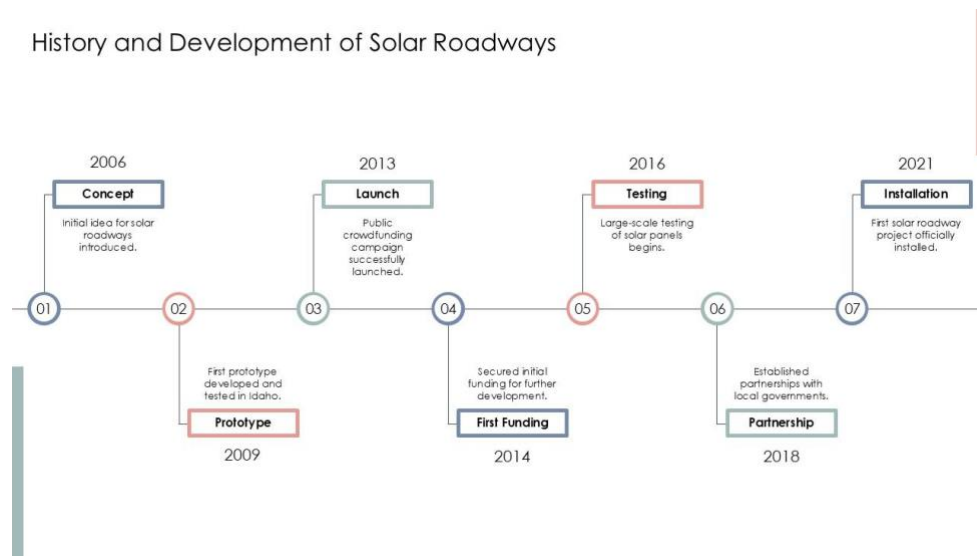


Figure 4: History of Solar Roadways

Physical architecture

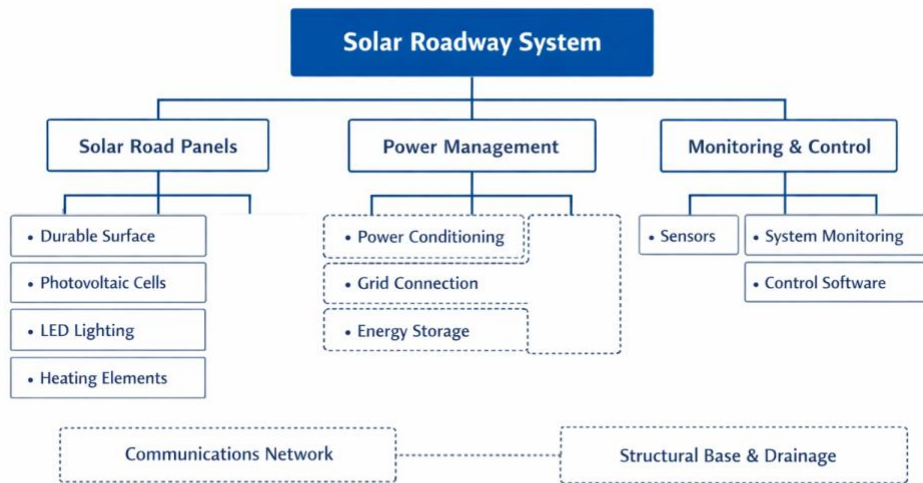


Figure 5: Physical Architecture Hierarchical Block Diagram of Solar Roadway System

A solar roadway system is designed as a layered infrastructure platform that integrates energy generation, lighting, and environmental control directly into the pavement surface. At the system level, the roadway consists of an array of interconnected panels installed on a structural base and supported by power management and control electronics. Each panel operates as an independent module, allowing for localized replacement which creates long-term infrastructure reliability and lifecycle cost requirements (U.S. Department of Transportation, 2022).

The uppermost layer of each panel is a load bearing, textured glass surface engineered to withstand vehicle loads while providing sufficient friction for safe travel. This surface must balance mechanical strength with optical transparency to ensure adequate sunlight reaches the photovoltaic layer beneath. Below the surface lies the photovoltaic component, which converts solar radiation into electrical energy. Since roadway panels lie flat, they receive less direct sunlight than tilted solar panels, so newer designs use more efficient solar cells to make up for this (International Energy Agency, 2020).

Embedded within the panel is a programmable lighting system composed of light-emitting diodes that can display lane markings, crosswalks, or warning signals. Unlike traditional painted markings, these LEDs allow roadway visuals to adapt dynamically to traffic patterns, construction zones, or emergency conditions. Heating elements are also

incorporated beneath the surface to reduce snow and ice accumulation. Environmental sensors monitor temperature and ambient conditions, enabling the heating system to activate only when necessary, which improves energy efficiency (Fay, Nazari, Jungwirth, & Muthumani, 2015).

The electricity produced by the photovoltaic layer flows through internal wiring that brings power from many panels together. This electricity can be used directly to run features such as lighting and surface heating, or it can be sent to an external system like a building or the electrical grid. At the same time, basic monitoring systems track how the panels are performing and alert operators when maintenance is needed. This level of oversight helps ensure the system remains safe, reliable, and functional in a roadway setting (Barbaresso, et al., 2014).

Beneath the panels is a sturdy base that supports traffic loads and provides space for wiring and water drainage. Additional equipment is needed at each site to move electricity where it is needed and to allow the system to be monitored and controlled. Together, these parts turn the roadway into more than just a driving surface by combining transportation, energy production, and basic communication functions. While pilot projects have shown that the concept can work, real-world testing is still being used to improve the design as the technology moves slowly toward broader use. (U.S. Department of Transportation, 2022; International Energy Agency, 2020).

Technical Specification

Solar roadway systems using modular photovoltaic pavement tiles shall meet the following technical requirements to be viable for pedestrian, vehicular, and environmental infrastructure applications.

Solar roadway systems shall:

I. Convert incident solar radiation into electrical energy with a minimum photovoltaic module efficiency of 15% under standard test conditions, consistent with demonstrated performance of photovoltaic pavement systems and commercial solar technologies (Shekhar, et al., 2018; National Renewable Energy Lab, 2024)

II. Withstand pedestrian and light maintenance vehicle loading in accordance with applicable structural performance standards used in transportation infrastructure, without structural or electrical failure (AASHTO, 2020; National Renewable Energy Lab, 2024).

III. Maintain a surface coefficient of friction equal to or greater than that of conventional concrete or asphalt pavement in both wet and dry conditions to ensure pedestrian safety (ASTM International, 2021).

IV. Operate reliably within an ambient temperature range of $-40\text{ }^{\circ}\text{C}$ to $60\text{ }^{\circ}\text{C}$, including resistance to freeze–thaw cycling and thermal expansion effects common in outdoor environments (Abdualla, et al., 2023).

V. Utilize a high-strength, impact-resistant, and textured surface layer that provides long-term durability while allowing sufficient light transmission to underlying photovoltaic cells (Shekhar, et al., 2018; Marsh, 2023).

VI. Be designed as modular, self-contained pavement units that can be individually removed and replaced without requiring full reconstruction of the surrounding walking surface (Shekhar, et al., 2018; Marsh, 2023).

VII. Include sealed and weather-resistant electrical components suitable for long-term outdoor deployment to minimize moisture ingress and electrical degradation (Abdualla, et al., 2023).

VIII. Incorporate integrated power electronics capable of conditioning generated electricity for safe local use or grid interconnection (National Renewable Energy Lab, 2024).

IX. Provide electrical isolation, grounding, and fault protection mechanisms to ensure safety for pedestrians, maintenance personnel, and nearby infrastructure (AASHTO, 2020).

X. Include an integrated or locally connected battery energy storage system capable of supplying backup power during grid outages to maintain critical system functions such as lighting and safety signaling (Marsh, 2023).

XI. Ensure the battery system provides a minimum of 8 hours of backup power for essential pedestrian safety features under nominal operating conditions, supporting infrastructure resilience and public safety (Marsh, 2023).

XII. Include integrated thermal management or heating functionality to mitigate snow and ice accumulation and improve winter walkability in cold climates (Abdualla, et al., 2023).

XIII. Achieve a minimum operational lifespan of 20 years, with total power output degradation not exceeding 20% over the system lifetime, consistent with established photovoltaic performance trends (Jordan & Kurtz, 2013).

XIV. Be installable using standard pedestrian infrastructure construction practices, enabling adoption without specialized or proprietary installation methods (Marsh, 2023).

Solar roadway systems should:

- I. Achieve photovoltaic conversion efficiencies exceeding 20% as solar cell technologies continue to advance (National Renewable Energy Lab, 2024).
- II. Incorporate embedded lighting elements to support enhanced visibility, wayfinding, and pedestrian safety (Marsh, 2023).
- III. Support integration of sensors or monitoring systems for performance tracking, maintenance planning, or pedestrian traffic analysis (Shekhar, et al., 2018).
- IV. Be cost-competitive on a lifecycle basis with conventional pedestrian pavement combined with separately installed solar generation systems (Marsh, 2023).
- V. Use durable, recyclable, or low-carbon materials where feasible to reduce environmental impact across the system lifecycle.
- VI. Allow for scalable deployment, from small demonstration walkways to larger pedestrian networks, without fundamental redesign of the system architecture.

Comparison with alternative technological approaches

The two leading alternative technological approaches to photovoltaic pavement in pedestrian infrastructure are solar canopies and traditional solar panels, which represent the incumbent technology. Solar canopies consist of elevated photovoltaic panels installed over sidewalks, bike paths, or adjacent pedestrian areas, allowing for high-efficiency energy generation without structural integration into the pavement itself. Conventional roadside solar panels are widely installed alongside roads or city infrastructure. They prioritize energy generation performance, scalability, and maturity. In contrast, photovoltaic pavement integrates energy generation directly into walking surfaces, enabling electricity production while simultaneously serving as pedestrian infrastructure.

One critical innovation success factor for evaluating these approaches is extensibility, defined as the ability of the technology to support additional functions beyond electricity generation. As shown in Figure 6, photovoltaic pavement scores highest in extensibility because it can incorporate complementary features such as snow and ice mitigation, embedded lighting for traffic control, and other smart city technologies. Solar canopies and roadside solar panels, while effective at generating power, are largely limited to a single function and therefore received lower extensibility scores. This factor was given a high importance rating due to increasing demand for multi-functional, smart infrastructure in dense pedestrian environments where space and budgets are constrained.

Another key success factor is pedestrian advantages, defined as the extent to which the technology directly improves pedestrian safety, experience, and operations. Photovoltaic pavement offers clear advantages in this category by providing underfoot benefits such as improved winter safety, dynamic lighting, and enhanced walkability.

Solar canopies offer limited indirect benefits such as shading, while roadside solar panels provide no direct pedestrian interaction. Pedestrian advantages were assigned a high importance rating, as technologies deployed in walkways and shared-use paths must deliver value beyond energy generation alone. This advantage is our key advantage over the competitors and incumbent technology. While photovoltaic pavement underperforms the alternatives in scalability and maturity, these user-centered benefits suggest that it is best suited for niche, high-visibility pedestrian environments rather than large-scale energy production.

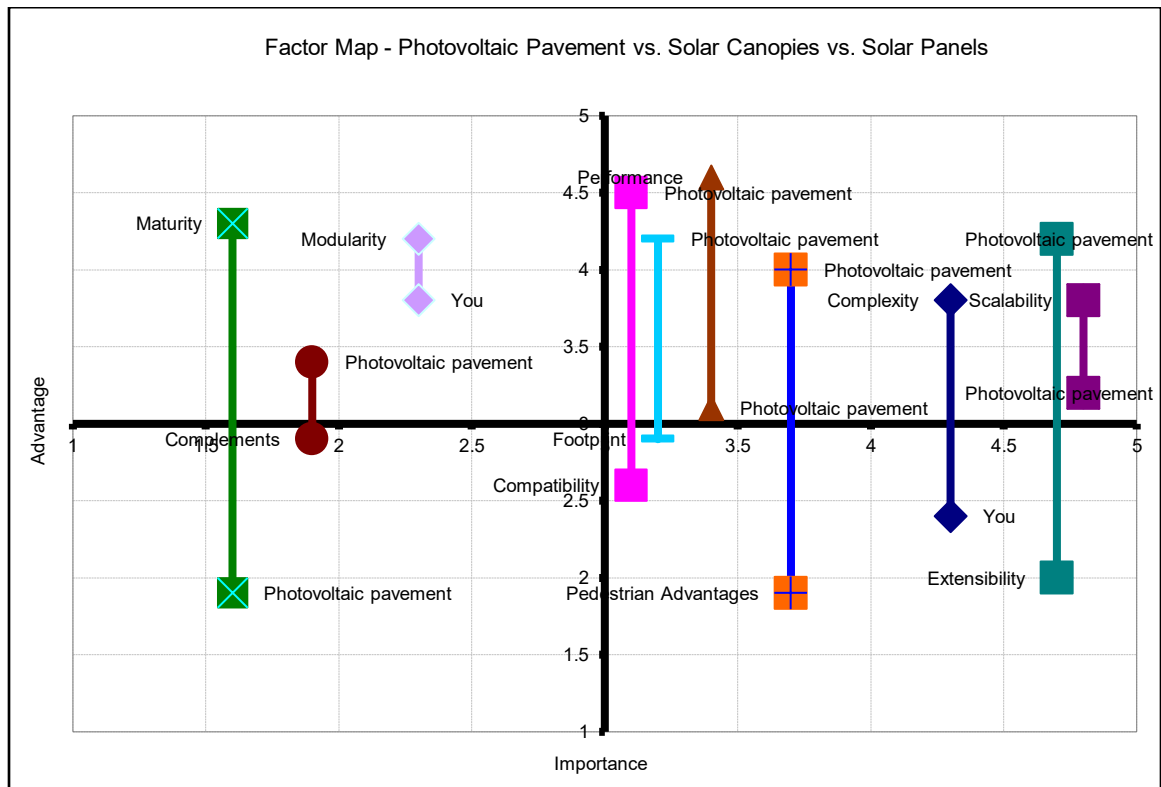


Figure 6: Importance-Advantage Chart for Photovoltaic Pavement vs. Solar Canopies vs. Solar Panels

Pipeline (follow-on) technologies

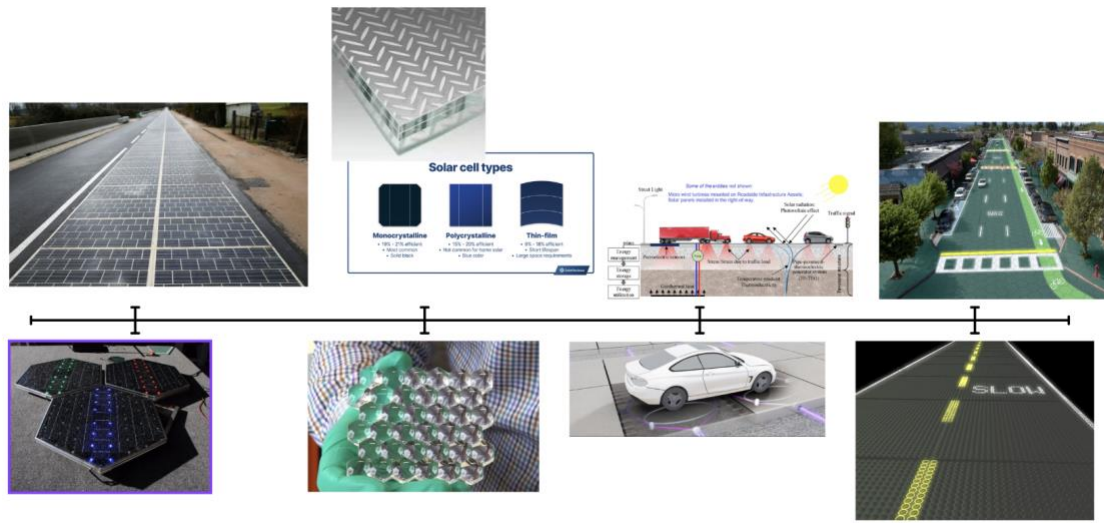


Figure 7: Potential timeline of photovoltaic pavement starting at current technology.

Since solar roadways are still a developing technology, they will need a clear pipeline of follow-on improvements in order to become practical and competitive over time. The first stage of this pipeline focuses on improving the current solar pavement design rather than completely reinventing it. This includes using more efficient solar cells, stronger and safer surface materials, and better protective coatings that improve traction and reduce wear. Making the panels more modular is especially important, since being able to replace individual panels instead of entire roadway sections would significantly lower maintenance costs and downtime.

As the technology improves, later versions of solar roadways could begin to include additional built-in features beyond basic energy generation. These include sensors to monitor traffic and roadway conditions, heating elements to prevent ice buildup in cold weather, and better integration with local energy storage systems. In the long term, solar roadways could evolve into a form of smart infrastructure, supporting features like adaptive lighting or limited wireless charging for electric vehicles in low-speed areas. These advanced applications would most likely be introduced in controlled environments such as campuses or private developments before being considered for wider use.

Implications for your technology strategy

The pipeline analysis highlights several key technical challenges that must be addressed before solar roadways can be successfully introduced as a minimum viable product. The most immediate challenge is durability, since the roadway surface must withstand traffic loads, weather exposure, and long-term wear while still remaining safe for vehicles and pedestrians. Another major issue is low energy efficiency compared to

traditional solar panels, which limits the technology's overall value and makes it harder to justify higher costs.

Because of these challenges, the initial technology strategy should focus on getting the basics right first. Early development efforts should prioritize durability, safety, and basic energy generation in low-traffic environments, rather than trying to include every possible advanced feature. Maintenance and repair must also be simplified through modular designs to make the system more realistic for real-world use. More complex features such as embedded sensors, heating systems, and smart grid integration should be treated as future upgrades, once the core technology has been proven. This step-by-step approach reduces technical risk and allows solar roadways to improve gradually as more data and experience are collected.

Competitive Environment

Factors to Consider

Top competitors and basis of competition

Solar Roadways operates at the intersection of renewable energy and transportation infrastructure, which places it in competition with both solar pavement developers and alternative smart road technology providers. One of the most significant direct competitors is Colas through its Wattway product. Colas is a global transportation infrastructure company that developed Wattway as a photovoltaic layer designed to be installed directly onto existing road surfaces. The company markets Wattway as a solution that integrates into traditional paving workflows while generating renewable electricity (Colas, Colas is revolutionizing roads with Wattway Solar Roads [Press Release], 2015). Additionally, industry reporting has highlighted efforts by Colas and research partners to secure relevant photovoltaic certifications, reinforcing its positioning as a credible infrastructure-grade solution (Bellini, 2024). Since Colas already maintains long-standing relationships with transportation agencies and construction partners, it competes strongly on reliability claims and institutional credibility.

Another direct competitor in the solar pavement space is SolaRoad, a Netherlands-based initiative associated with applied research organizations such as TNO. SolaRoad developed and tested photovoltaic bike paths and road surfaces in real-world conditions, publicly reporting performance outcomes such as annual energy yield per square meter and durability observations (EARTO, n.d.). Early pilot coverage documented both energy production results and technical challenges related to surface wear and maintenance, which have shaped industry perceptions of solar pavement feasibility (Cooper, 2015). Even where limitations were encountered, the SolaRoad project helped define the benchmarks by which future solar roadway technologies are evaluated. As a result, Solar Roadways must compete not only against SolaRoad's technology, but also against the expectations established by its published performance data and pilot outcomes.

In addition to direct solar pavement developers, Solar Roadways faces indirect competition from companies pursuing alternative smart road technologies. Electreon represents a prominent example. The company develops wireless electric road systems that allow electric vehicles to charge dynamically while driving. Although this technology does not involve photovoltaic generation, it competes for the same infrastructure modernization funding and strategic attention from transportation agencies. Public reporting has highlighted Electreon's pilot projects and partnerships, including high-profile demonstrations in the United States, which position the company as a leader in road-based electrification solutions (Maynard, 2023; Descant, 2022). Since many transportation agencies prioritize electrification and fleet efficiency, Electreon competes by offering a solution aligned with those policy objectives.

The basis of competition in this industry centers on engineering durability, cost effectiveness, and institutional credibility. Solar Roadways must demonstrate that its surface can withstand traffic loads and environmental stress while maintaining traction and structural integrity. It must also show that energy output is sufficient to justify installation and maintenance costs relative to conventional solar installations. Competitors such as Colas and Electreon benefit from established infrastructure relationships and documented field deployments, which heightens the competitive pressure on emerging entrants in the solar roadway space.

Competitive Landscape

Building on this competitive landscape, the primary value disciplines shaping competition in the solar roadway industry are product leadership, operational leadership, and customer intimacy. Figure 8 maps Solar Roadways, Colas (Wattway), and Electreon across these value disciplines using a center-of-gravity framework.

In terms of product leadership, Solar Roadways received a score of 4. The company has positioned itself as a innovative solution by integrating photovoltaic generation, pedestrian signaling, and heating systems into a modular system. This multifunctional design differentiates it from competitors by combining energy production, smart signaling, and potential safety enhancements into a single infrastructure unit. Colas received a slightly lower score of 3 because Wattway focuses more narrowly on applying a photovoltaic layer to existing pavement rather than replacing the roadway system itself. Electreon scored a 4 in product leadership, as its wireless charging technology represents a technically sophisticated and novel approach to electrified infrastructure, even though it does not generate solar power directly.

With respect to operational leadership, Colas scored the highest at 5. As a global transportation infrastructure firm with established procurement channels, construction partnerships, and public-sector credibility, Colas benefits from decades of operational experience delivering large-scale roadway projects. This provides a significant advantage in deployment reliability and institutional trust. Electreon received a 4, reflecting its growing portfolio of pilot projects and government partnerships, though it does not yet operate at the same scale as a multinational contractor like Colas. Solar Roadways received a 2 in this category, primarily because it remains an emerging company without comparable large-scale field deployments or long-standing infrastructure contracts. Operational maturity and proven scalability remain key hurdles.

Finally, in terms of customer intimacy, both Colas and Electreon scored 4, as each company works directly with transportation agencies and aligns its solutions with policy priorities such as electrification, grid integration, and infrastructure modernization. Their experience navigating public procurement processes strengthens their ability to tailor solutions to institutional buyers. Solar Roadways received a 2 in this dimension. While the company has generated public interest and community-level pilot discussions, it does not yet demonstrate the same depth of ongoing institutional partnerships or long-term agency collaboration as its competitors.

Overall, the center-of-gravity mapping shows that Solar Roadways currently competes most strongly on product vision and conceptual innovation, while Colas dominates on operational execution and institutional credibility. Electreon occupies a relatively balanced position with strong technological sophistication and growing deployment credibility. For Solar Roadways to shift its competitive center of gravity closer to the industry leaders, it must improve operational scalability and deepen strategic partnerships with infrastructure agencies to complement its innovative product positioning.

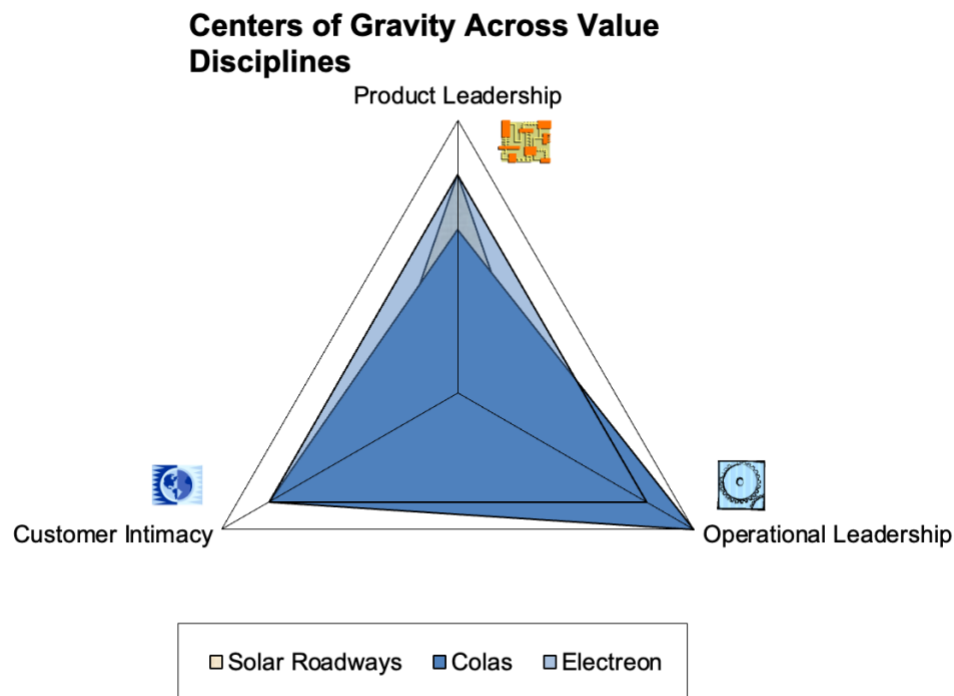


Figure 8: Center of Gravity Chart for Competitors

Market share and cost structure are central competitive factors in the solar roadway industry. Colas, through Wattway, benefits from the broadest effective market access due to its integration within a global transportation infrastructure firm, allowing it to leverage established procurement channels and construction workflows to lower deployment costs (Colas, Colas is revolutionizing roads with Wattway Solar Roads [Press Release], 2015; Bellini, 2024). Electreon holds a narrower share but is gaining traction through high-visibility electrification pilots, positioning itself within government EV

infrastructure funding despite relatively high capital intensity (Maynard, 2023; Descant, 2022). In contrast, Solar Roadways currently operates with a limited initial market share and high per-unit production costs due to the technical complexity of integrating photovoltaic cells, LEDs, heating systems, and structural materials into modular panels. As a result, Solar Roadways falls within the differentiation focus quadrant of the cost-share matrix.

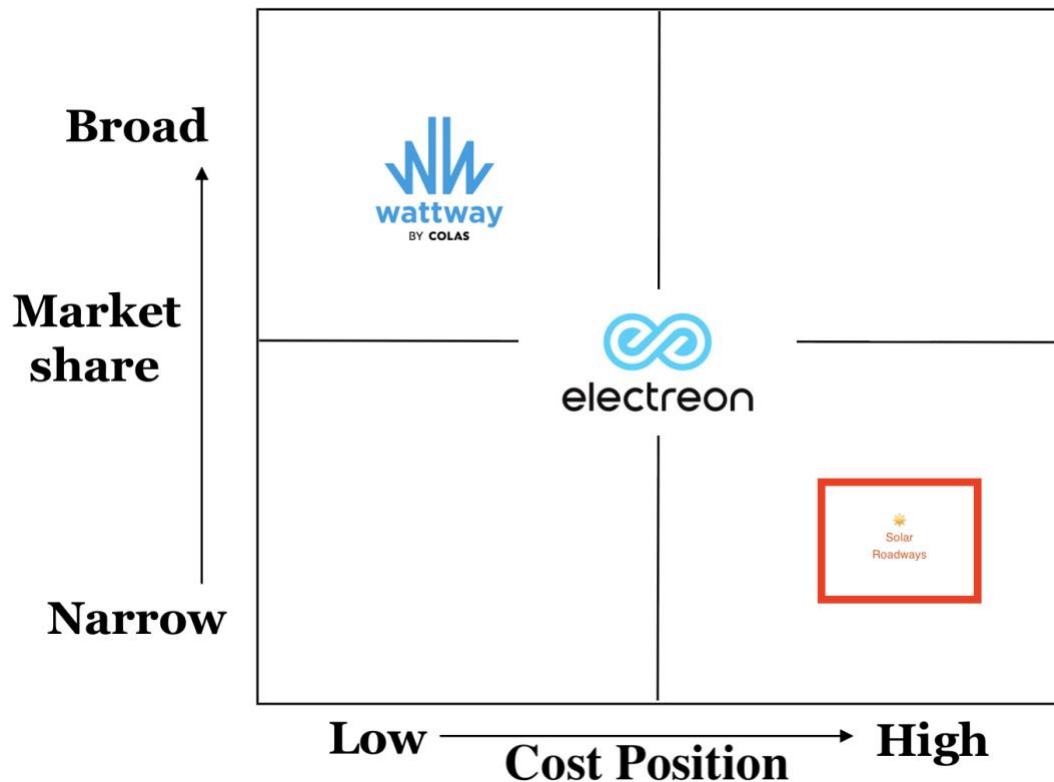


Figure 9: Cost-Share Matrix.

Proprietary Position

Solar Roadways benefits from a relatively strong proprietary position due to the technical integration required to embed photovoltaic generation, electronics, and structural load-bearing materials into a roadway-grade surface. Achieving durability, traction compliance, and electrical reliability simultaneously presents nontrivial engineering barriers. While competitors such as Colas apply photovoltaic layers to existing surfaces and Electreon embeds inductive charging coils beneath pavement, Solar Roadways' fully modular surface architecture increases imitation complexity. However, unlike Colas, Solar Roadways does not benefit from deep institutional lock-in within transportation procurement ecosystems.

Speed

As a smaller and more specialized product, Solar Roadways has greater strategic agility than larger incumbents. Colas operates within established infrastructure processes that may slow rapid design iteration, and Electreon must coordinate closely with automotive manufacturers and regulators. Solar Roadways, by contrast, can refine panel designs,

target niche applications such as pedestrian walkways or campuses, and pivot its deployment strategy more quickly in response to pilot performance data. This agility may partially offset its scale disadvantages.

Product Line Breadth

Colas maintains the broadest product portfolio, spanning traditional paving services, infrastructure engineering, and renewable surface solutions. Electreon focuses primarily on dynamic charging systems but operates within the broader EV ecosystem. Solar Roadways has the narrowest product line, concentrating almost exclusively on modular solar pavement panels. While this focused strategy allows technical refinement, it limits complementary revenue streams compared to competitors with diversified infrastructure portfolios.

Scale and Scope Economies

Scale economies currently favor Colas most strongly due to its global construction operations and optimized procurement networks. Electreon benefits from scaling within the EV infrastructure segment but still faces high capital intensity. Solar Roadways, as an emerging technology provider, would initially experience high per-unit production costs and limited manufacturing efficiencies. However, the modular design of its panels offers potential scope economies, as the same core unit could be deployed across sidewalks, bike paths, parking lots, and private roadways with relatively minor modification. This cross-application flexibility could improve cost amortization over time.

Experience Curve Effects

Colas possesses the most extensive experience curve advantage, having delivered roadway infrastructure projects for decades. Electreon is building operational experience through pilot deployments. Solar Roadways, by comparison, remains early on the learning curve. Until significant field installations accumulate performance data and process optimization insights, its cost structure will remain comparatively high.

Design Dominance and Complements

It remains uncertain which smart roadway technology will achieve long-term design dominance. Traditional asphalt paired with standalone solar installations remains the benchmark for cost efficiency, while Colas emphasizes integration compatibility and Electreon aligns with EV electrification policy goals. Solar Roadways differentiates itself through multifunctionality, but must demonstrate sustained durability and energy output to validate its design superiority. In terms of complements, Colas benefits from deep contractor networks and infrastructure partnerships, while Electreon aligns with automotive manufacturers and grid operators. Solar Roadways must actively cultivate complementary relationships with municipalities, campus facilities managers, and renewable energy integrators to strengthen its competitive position.

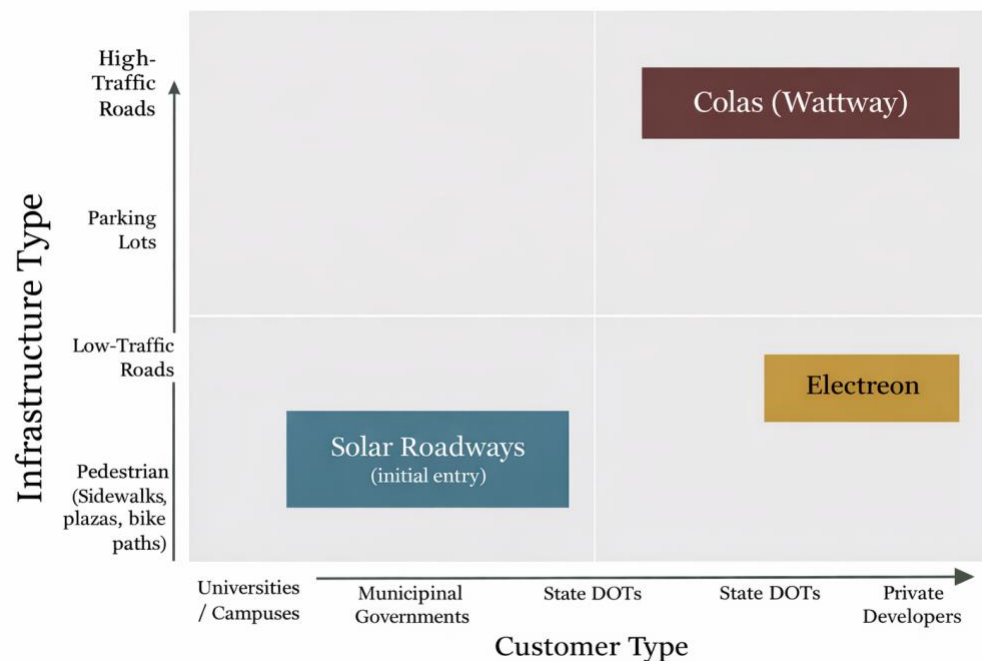


Figure 10: Dimensions of Competitive Landscape.

Customer Value Proposition

For campuses and infrastructure developers seeking sustainable and resilient public infrastructure, solar roadways provide a multifunctional roadway system that generates renewable energy while improving safety and operational efficiency. Unlike traditional roads and standalone solar installations, solar roadways integrate energy production, lighting, heating, and sensing directly into existing transportation surfaces, allowing customers to create value from infrastructure they already must build and maintain. By transforming roadways from passive cost centers into active, value-generating assets, solar roadways enable customers to meet climate goals, enhance public safety, and modernize infrastructure without requiring additional land use or separate energy systems.

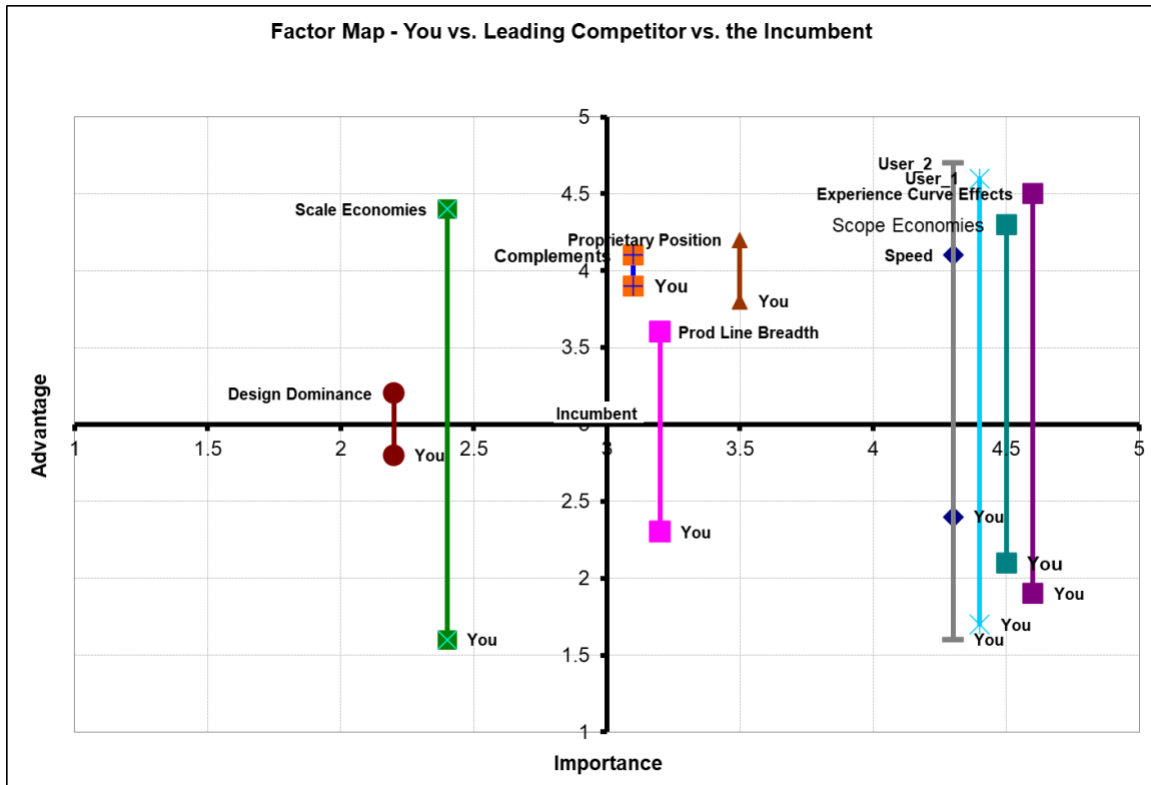


Figure 11: Factor Map Comparing Solar Roadways, Leading Competitor, and Incumbent Infrastructure

Projected market share

Figure 12 illustrates projected market share redistribution between 2026 and 2033. In 2026, the incumbent is assumed to control nearly 100% of the market, which reflects the dominance of the established infrastructure providers with existing procurement relationships and already-active roadway systems. Over time, market share is expected to shift as smart roadways and solar-integrated technologies become more technically competent and acceptance, and achieves broader institutional adoption.

Our company is projected to begin capturing measurable market share in 2027 following early pilot deployments. Growth would be expected to be incremental, reaching approximately 15–20% by 2030 and approximately 25% by 2033. This projection reflects a phased commercialization strategy focused initially on niche, small-scale applications such as bike paths, pedestrian zones, and small-town infrastructure before expanding to larger roadways.

Competitors such as Colas (Wattway) and Electreon are also projected to gain share due to established credibility and currently active pilot programs (Colas, n.d.; Time, 2023; Government Technology, 2023). Additional companies, including solar pavement developers such as SolaRoad, will contribute to overall market diversification (EARTO, n.d.; Cooper, 2015).

The projected 25% share by 2033 represents a strategic but realistic growth target given the capital intensity, public establishment cycles, and engineering performance requirements in transportation infrastructure markets.

Projected Market Share vs. Incumbent, Key Competitors

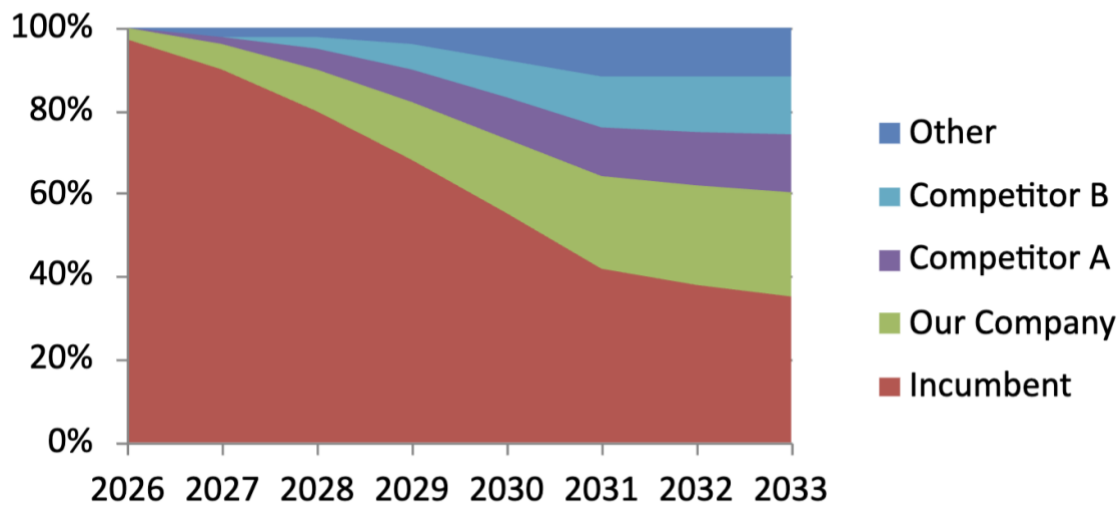


Figure 12: Projected Market Share vs. Incumbent

Host Company Identification and Assessment

Capabilities required

When evaluating potential host companies for solar roadway technology, several key capabilities are necessary to successfully develop and implement the product. First, the host company must be in a strong financial position. Solar roadways require significant upfront investment in research and development, prototyping, materials testing, and pilot installations. Because the technology currently faces cost and durability challenges compared to traditional asphalt and concrete, the host company must be able to support continued R&D efforts without expecting immediate profitability. Long development timelines and uncertain returns make financial stability one of the most important characteristics of a suitable host.

Another essential capability is expertise in large-scale infrastructure or construction projects. Solar roadways combine elements of civil engineering, materials science, and renewable energy systems, meaning the host must have experience managing complex, multidisciplinary operations. Companies that already specialize in infrastructure development, commercial property construction, or renewable energy integration would be better prepared to handle installation logistics, maintenance planning, and regulatory compliance. Because the technology will likely be introduced

first in private developments such as parking lots and pedestrian pathways, the host should also have strong relationships with commercial developers and property owners.

In addition, the ideal host company should demonstrate a commitment to sustainability and innovation. Since solar roadways align with environmental goals and smart infrastructure initiatives, the host should already have a strategic focus on renewable energy, green building practices, or ESG (environmental, social, and governance) objectives. This alignment helps ensure that the company's customer base overlaps with early adopters who value sustainable solutions and may be willing to pay a premium for innovative infrastructure. A company that actively markets itself as forward-thinking and environmentally responsible would benefit more from integrating solar roadway technology into its portfolio.

Finally, operational excellence is critical. Because solar roadways must meet strict safety and performance standards, the host company must already demonstrate high-quality execution in its existing operations. Durability, surface traction, electrical reliability, and maintenance efficiency are all technical challenges that require careful implementation. A host company that dominates its current operations will be better positioned to refine the technology, manage risk, and gradually scale adoption. Overall, financial strength, infrastructure expertise, sustainability alignment, and operational excellence are the core capabilities required for a company to successfully adopt and advance solar roadway technology.

Candidate host companies

In evaluating potential host companies for solar roadway implementation, our team focused on firms that already build and manage large transportation systems over long periods of time. Since solar roadways combine traditional pavement engineering with solar panels and grid connections, the host company must be able to design and deliver complex road projects while also integrating electrical components into those projects. The host company must also invest consistently in research and have the financial strength to support multiple rounds of testing. With these factors in mind, Colas, VINCI, and Ferrovial emerged as strong candidates.

To compare them fairly, we created a weighted scoring model in Excel using publicly available information. Each company was scored from one to ten across the required capabilities, and we gave more weight to research funding and roadway performance because new infrastructure technologies require repeated testing and dependable results in real-world conditions. After applying the weights, Colas received the highest overall score, with VINCI close behind and Ferrovial slightly lower. The differences were not large, but they reflected which companies had more direct experience and technical focus related to solar road systems.

Colas is a global road construction company that builds and maintains highways, streets, and transportation networks in many countries. Its core business centers on pavement materials and roadway engineering, which closely aligns with the foundation of solar roadway technology. Colas stands out because it has already developed and tested a solar pavement product called Wattway. The company has installed this technology and continues to improve it, treating solar roads as a serious engineering project rather than just an environmental concept (Colas, Wattway by Colas, 2023). Through these trials, Colas has gained experience with durability and electricity output. Early solar road projects in France faced issues with efficiency and noise, showing that the concept still requires refinement (Grossman, 2019). Colas has the scale and resources to continue improving the technology instead of stopping after early setbacks.

VINCI is one of the largest infrastructure companies in the world and builds roads, bridges, tunnels, and airports across many countries. Through its subsidiary Eurovia, it focuses heavily on road construction and long-term maintenance. VINCI invests in improving road materials and works on projects that connect transportation and energy systems (VINCI, 2023). Its size and engineering strength mean it could scale a successful pilot quickly. However, compared to Colas, VINCI has shown less direct experience with roads that generate solar power, which makes it slightly less ready to commercialize solar roadways in the near future.

Ferrovial is a global infrastructure company that builds and operates highways and airports and often manages these projects for decades. This long-term operating experience shows that it understands how to maintain large transportation assets over their full life cycle (Ferrovial, 2023). The company promotes innovation and sustainability, but we found fewer examples of it directly testing solar pavement systems. Since it has less specific experience with roads that produce electricity, it received a slightly lower score in our evaluation.

Overall, Colas presents the strongest host for solar roadway deployment. Its deep experience in road construction, real-world testing of solar pavement technology, and continued investment in improving the design, gives it a practical advantage. Choosing a company that already works at the intersection of roadway engineering and renewable energy increases the likelihood that solar roadways can move from experimental pilots to viable infrastructure projects.

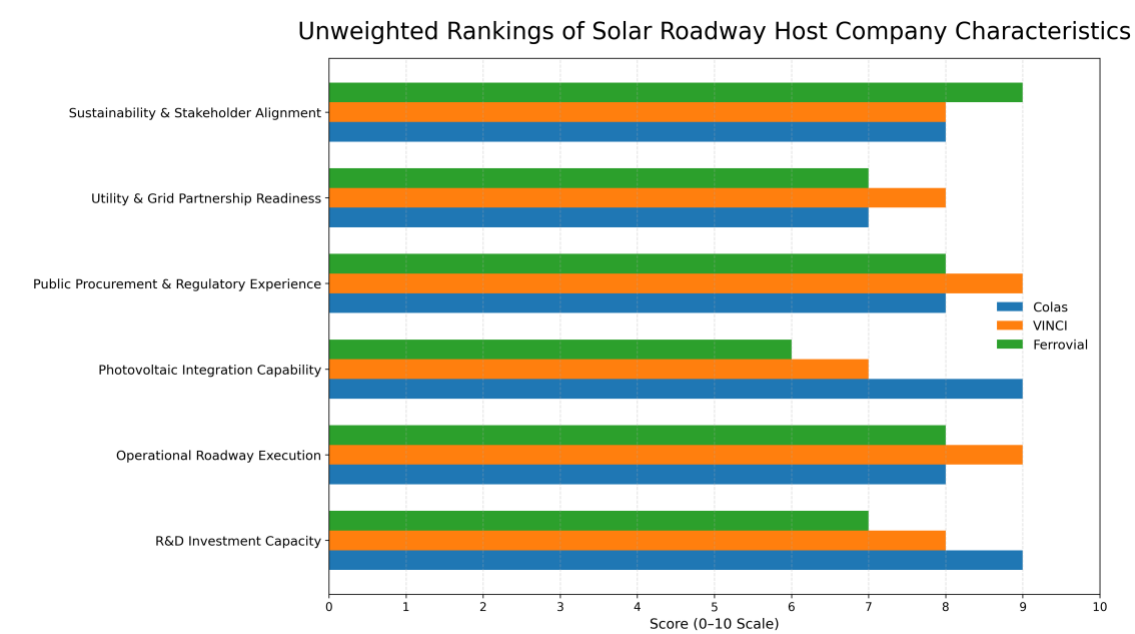


Figure 13: Unweighted Rankings of Host Company Characteristics

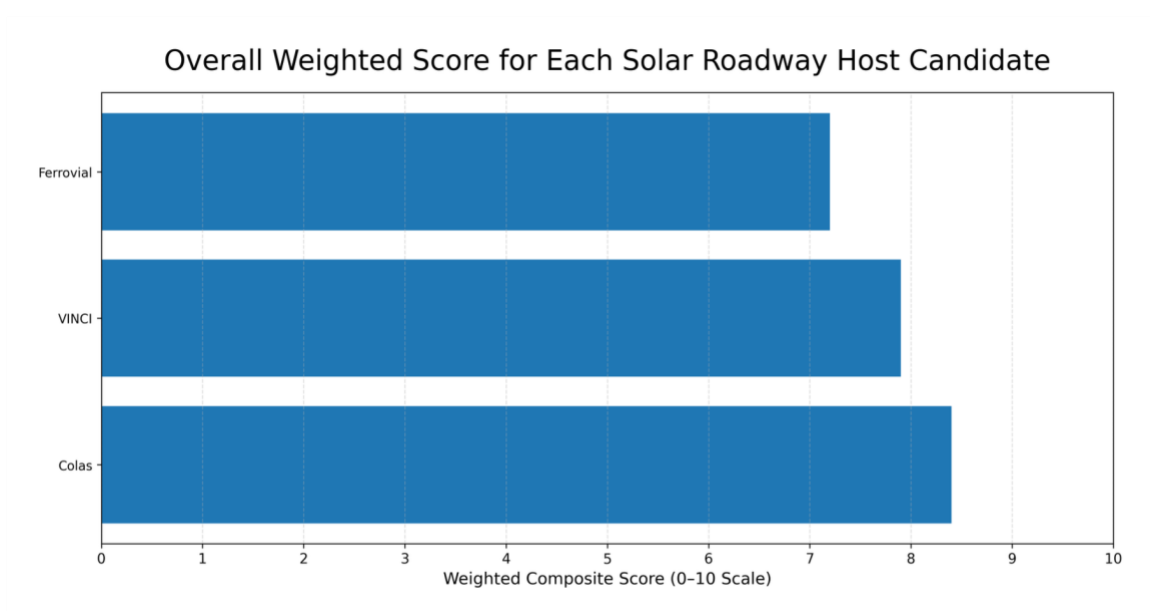


Figure 14: Overall Weighted Score for Host Candidates

Company's business, key customers or markets.

Colas Group is a global leader in the construction and maintenance of transport infrastructure, operating in around 50 countries across five continents and employing more than 63,900 *people* (Colas, 2024 Annual Report, 2025). ***The Group operates across three primary segments: roads, construction materials, and railways. Its roads business accounts for €14.4 billion of its €15.9 billion in total 2024 revenue*** (Colas, 2024 Annual

Report, 2025). *International operations represent 59% of total revenue, demonstrating its global outreach* (Colas, 2024 Annual Report, 2025). *This is a helpful quality when considering the acceptance of Solar Roadways throughout the United States, but also globally. Colas' customers range from central and local governments and public transport to private-sector developers, reflecting its strong position in public infrastructure markets. Strategically, Colas emphasizes innovation, sustainability, and low-carbon infrastructure through its ACT corporate plan and dedicated R&D entities such as the Core Center and Innovation Department* (Colas, 2024). *In 2024, 60% of its R&D projects were directly focused on reducing infrastructure carbon footprints or improving climate resilience* (Colas, 2024 Annual Report, 2025). *Notably, Colas continues to develop and commercialize Wattway, its photovoltaic road surfacing solution. This development shows both technical capability and organizational willingness to pursue energy-generating pavement technologies. Given its control of the road construction value chain (including 3,500 materials production units), strong public-sector relationships, and commitment to low-carbon innovation, Colas exhibits the R&D infrastructure, financial scale, and strategic alignment necessary to serve as a credible host company for further development and commercialization of advanced solar roadway systems.*

Analysts' views of company's strengths, weaknesses, and future prospects.

Recent business and technical publications show Colas Group as a financially stable infrastructure leader. Articles also mention its development of sustainable transport products. Coverage in PV Magazine highlights Colas' continued technical advancement of its photovoltaic pavement system, Wattway, noting the recent IEC certification of its PV road technology, which signals improved durability standards and grid compatibility (Bellini, 2024). Similarly, reporting from the Institut National de l'Énergie Solaire (INES) emphasizes that Wattway paving slabs have achieved formal compliance certification (Institute National de l'Énergie Solaire, 2024). This shows their technical credibility and commercial readiness. Articles from the broader energy sector report Colas' strategic partnerships, particularly its renewed alliance with TotalEnergies aimed at reducing the carbon footprint of construction operations, which reinforces the company's positioning as a sustainability-driven contractor (Colas, 2025).

However, while technical publications acknowledge innovation momentum, they frame photovoltaic roadways as a niche complement rather than a near-term substitute for conventional solar installations. PV Magazine notes that horizontal solar installations face inherent efficiency and cost challenges compared to rooftop systems, suggesting commercialization will likely focus on targeted urban or demonstration applications rather than highway-scale deployment (Bellini, 2024). This is the same issue we identified early with our product. This contrasts slightly with Colas' own framing of Wattway as part of a broader transformation of roads into energy-generating infrastructure assets (Colas, 2015). Overall though, external coverage aligns closely with the strategy discussed in Colas' 2024 Annual Report, which emphasizes decarbonization, innovation, and infrastructure resilience as core pillars of future growth (Colas, 2024 Annual Report, 2025). The primary difference between external sources and Colas lies in commercialization expectations. External observers recognize strong R&D and

certification progress but remain less excited regarding large-scale economic viability. On the other hand, Colas presents photovoltaic pavement as a scalable contributor to its low-carbon infrastructure roadmap. For solar roadway technology, this suggests alignment with Colas' sustainability strategy, though near-term success will likely depend on selective, high-value use cases rather than broad market disruption.

Company's technical strengths

Centers of Gravity Across Value Disciplines

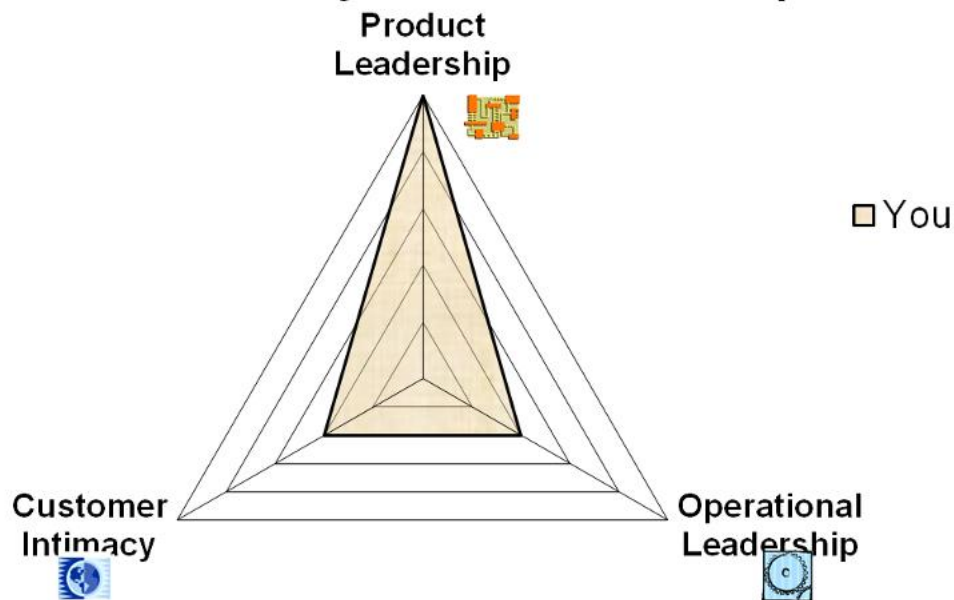


Figure 15: Center of Gravity for Solar Roadways Company

Solar Roadways' core competency lies in integrated product innovation, positioning the company within the Product Leadership value discipline. The company develops modular, load-bearing solar pavement panels that combine photovoltaic generation, LED lighting, and heating elements into a single infrastructure platform (Solar Roadways, The Solar Road Panel, n.d.). This multifunctional integration differentiates Solar Roadways from conventional solar energy firms and traditional asphalt infrastructure providers.

The company's primary technical strength is the engineering integration of energy generation with structural roadway requirements. Unlike traditional photovoltaic systems installed on rooftops or ground-mounted arrays, Solar Roadways embeds solar cells within tempered glass panels designed to withstand vehicle loads (Solar Roadways, The Solar Road Panel, n.d.). Company materials emphasize durability testing, surface traction, embedded circuitry, and heating capabilities, indicating a strong research and development orientation. This emphasis on technological differentiation aligns with the

Product Leadership discipline described by Treacy and Wiersema (1995), in which firms compete primarily through innovation and performance rather than operational efficiency.

Solar Roadways also demonstrates systems-level innovation. In addition to generating electricity, the panels incorporate programmable LED lighting and snow-melting heating elements (Solar Roadways, The Solar Road Panel, n.d.). This integrated smart infrastructure approach positions the company as a technology developer rather than a conventional construction contractor. By comparison, competitors such as Colas' Wattway focus primarily on applying photovoltaic surfacing to existing road infrastructure (Colas, Wattway by Colas, 2023), while firms such as Electreon focus on electrified roadway charging systems (Maynard, 2023). Solar Roadways' differentiation lies in combining multiple infrastructure functions within a single modular product.

Based on this analysis, Solar Roadways' center of gravity is clearly Product Leadership. The company does not currently demonstrate strengths in Operational Leadership, as it lacks the large-scale manufacturing networks and established public procurement relationships of global infrastructure firms such as Colas (Colas, 2024 Annual Report, 2025). Nor does it primarily compete on Customer Intimacy, as it does not yet maintain extensive service or partner ecosystems. Instead, its competitive advantage depends on continued technological development, intellectual property protection, and performance validation.

To successfully commercialize its technology, Solar Roadways must maintain its innovation focus while developing strategic partnerships to strengthen operational capacity. Overall, available evidence suggests the company competes primarily through technological differentiation and integrated product innovation.

Strategic implications

Based on our earlier evaluation, Colas emerges as the strongest proposed host company for solar roadway implementation because its core competencies align directly with the technical demands of this technology. Solar roadways must withstand heavy traffic loads, temperature fluctuations, moisture, and long term surface wear while maintaining consistent electricity output. Colas' core business centers on pavement engineering and roadway materials, which gives it a strong center of gravity in exactly the area solar roadways depend on most. Unlike firms that approach solar roads primarily from an energy perspective, Colas approaches them first as a road engineering challenge. Its development and testing of the Wattway solar pavement system demonstrates that it has already confronted real world durability and performance issues rather than treating the concept as purely theoretical (Colas, Wattway by Colas, 2023).

A key strength of Colas is that it has already invested in repeated field trials. Early Wattway installations in France revealed issues related to surface noise and long-term

resilience under traffic loads (Grossman, 2019). While these setbacks exposed technological weaknesses, they also show that Colas understands the experimental nature of infrastructure innovation and has the financial strength to continue refining the system. Infrastructure technologies often require multiple cycles of prototyping and adjustment before reaching commercial viability. Since Colas operates globally and maintains long term roadway contracts, it has both the testing environments and capital base necessary to sustain this iterative development process. Its strong relationships with governments make it easier for the company to get approval for pilot projects and testing sites, which is especially important for roads that also produce and connect electricity to the power grid.

Even with these strengths, there are still major challenges. Solar roadways are currently less efficient and more expensive than regular rooftop or large solar farms. If the technology does not improve, Colas could spend money on something that cannot compete with cheaper renewable options. Another challenge is expanding from small test projects to large scale use. Small pilots may work, but using solar roads everywhere would require consistent production and clear long term maintenance plans. Since Colas mainly works with traditional asphalt, there could also be internal pushbacks. Some teams may prefer sticking with proven pavement methods instead of shifting toward a newer and more experimental solar surface.

The company will also face broader outside pressures. Factors such as changes in government policy and how the public views the value of solar roadways will strongly influence whether the technology is adopted. Solar roadways must compete with traditional asphalt and also with less expensive solar panels that can be installed alongside highways instead of embedded within them. If economic conditions weaken or infrastructure budgets become tighter, experimental technologies are often delayed. At the same time, if governments continue to expand carbon reduction goals and invest in visible sustainability projects, interest in innovative infrastructure like solar roads could grow.

If Colas does not yet have all the technical capabilities required for large scale solar roadway deployment, it can strengthen its position by forming closer partnerships with solar panel manufacturers and smart grid technology companies. Colas is highly skilled in building durable pavement systems, but maximizing electricity production and ensuring connection to the power grid may require deeper expertise in photovoltaic engineering. By creating joint ventures or investing in smaller clean energy firms, Colas could gain this knowledge without shifting away from its core identity as a roadway engineering company. This would allow it to fill gaps in electrical capabilities while continuing to build on its strength in materials science and pavement durability.

Overall, Colas holds a meaningful advantage because of its experience in pavement engineering, its real world testing of solar roadway systems, and its long term management of transportation networks. Compared to competitors such as Vinci SA and Ferrovia, Colas is more directly aligned with the technical demands of solar road technology. Its main challenge will be proving that solar roadways can meet durability

standards and remain cost effective at a large scale. If the company continues disciplined field testing while strengthening its ability to integrate electrical systems, it has the strongest foundation to move solar roadways from pilot projects to credible, long term infrastructure solutions.

Technology/Business Intelligence

Priority issues for intelligence collection

After reviewing Sections 1–4, several significant gaps remain regarding the technical feasibility, economic viability, and strategic positioning of Solar Roadways. While the concept demonstrates strong innovative appeal and sustainability potential, much of the available information is promotional in nature and lacks independent, long-term validation. To meaningfully assess the likelihood of commercialization and large-scale adoption, further clarification from industry experts is necessary in several high-priority areas.

One of the most critical gaps involves long-term durability under real-world traffic conditions. Although pilot installations demonstrate proof of concept, there is limited publicly available data on performance under sustained heavy axle loads, high-speed traffic, freeze–thaw cycles, and environmental wear. Understanding typical failure modes—such as surface cracking, delamination, or electrical degradation—is essential, as is comparing the expected lifespan of solar panels embedded in roadways to that of conventional asphalt or concrete. Without clear lifecycle performance data, it is difficult to determine maintenance frequency and overall cost effectiveness.

Economic competitiveness represents another major uncertainty. Solar Roadways must ultimately be evaluated against conventional solar photovoltaic systems in terms of levelized cost of energy (LCOE). The embedded electronics, modular construction, and specialized installation processes may significantly increase cost per kilowatt-hour relative to rooftop or utility-scale solar installations. An industry source could clarify whether a realistic pathway to cost reduction and scalability exists, and under what deployment conditions the system might approach economic parity with traditional alternatives.

System compatibility with existing infrastructure also requires further exploration. Large-scale implementation would necessitate distributed inverters, grid interconnection equipment, monitoring systems, and potentially storage integration. It remains unclear whether current municipal electrical grids could absorb decentralized roadway-generated electricity without substantial upgrades. Compatibility with existing power and transportation systems is a key determinant of adoption, particularly in risk-averse public infrastructure environments.

Regulatory and liability considerations present additional unknowns. Roadway materials must meet stringent federal and state transportation standards related to load-bearing

capacity, friction coefficients, safety, and structural reliability. Departments of Transportation operate under strict engineering guidelines, and any deviation from established materials introduces approval complexity. Furthermore, liability exposure in the event of panel failure causing vehicle damage or accidents could create insurance barriers that significantly impede adoption. Clarifying regulatory pathways and legal risk allocation is therefore a high-priority research need.

Operational and maintenance logistics also remain insufficiently defined. Although the modular design suggests ease of replacement, practical realities such as lane closures, repair time per panel, snow removal procedures, oil and debris resistance, and long-term reliability of embedded LEDs and heating elements must be assessed. Infrastructure systems succeed not only based on initial innovation but on manageable maintenance demands and predictable operating costs.

Finally, the initial market entry strategy warrants deeper investigation. Deploying Solar Roadways on major highways may represent an overly ambitious starting point for a breakthrough infrastructure technology. Alternative early adoption markets—such as parking lots, sidewalks, university campuses, military bases, or private developments—may offer lower regulatory barriers and more controlled testing environments. Understanding where the technology can gain early traction and build credibility is essential for developing a realistic scaling pathway.

Collectively, these research areas represent the most significant uncertainties affecting the commercial viability of Solar Roadways. Engaging directly with industry experts is necessary to obtain technical validation, economic benchmarking, and regulatory insight that cannot be fully derived from publicly available information.

Interview questions

1. Cost and Manufacturing
 - a. How does the projected cost of manufacturing modular solar roadway panels compare to traditional asphalt or concrete paving per square foot?
 - b. What are the most expensive components in a solar roadway panel system?
 - c. How much could costs decrease if production were scaled to commercial highway levels?
 - d. How different would installation and maintenance processes be compared to standard road construction methods?
2. Structural Safety
 - a. What load bearing standards must roadway materials meet for highway approval?
 - b. How would replacing asphalt with textured structural glass affect traction and skid resistance?
 - c. What long term durability concerns would you have regarding cracking, impact resistance, or fatigue under heavy traffic?
 - d. How might extreme temperature fluctuations affect the structural integrity of embedded electronic components?

3. Energy Generation
 - a. How efficient would solar panels embedded in horizontal road surfaces be compared to tilted rooftop installations in energy production?
 - b. How much energy loss would you expect from dirt, shading by vehicles, or surface wear?
 - c. Would integrating heating elements significantly reduce net energy output?
 - d. In your opinion, could a roadway system realistically generate surplus energy for grid distribution?
4. Regulatory and Infrastructure Integration
 - e. What federal or state approval processes would this technology need to pass before public deployment?
 - f. What key regulatory barriers might delay adoption?
 - g. How complex would it be to connect distributed roadway panels to the existing electrical grid?
 - h. Would transportation agencies be open to piloting this type of infrastructure?
5. Market Feasibility and Adoption Timeline
 - i. What do you see as the primary barriers to large scale adoption of solar roadway systems?
 - j. Do you believe this technology could be commercially viable within the next 5 to 10 years?
 - k. Which early markets or pilot environments would be most realistic for initial implementation?
 - l. From your perspective, does the long term value of solar roadways justify the upfront investment?

Potential interview sources

The following individuals have been identified as potential interview sources based on their industry, qualifications, and ease-of-contact:

1. Thomas Rush
CEO – Electrovia (Nashville, TN)
Contact: thomas@electrovia.co

Thomas Rush is the CEO of Electrovia, a company in Nashville attempting to enable wireless charging for electric vehicles on roadways. His insight into energy systems, construction, and working with municipalities would be beneficial to developing our business strategy.

2. Josh D. Caldwell
Prof. & Researcher at Vanderbilt University
Contact: josh.caldwell@vanderbilt.edu

Dr. Caldwell is a researcher at Vanderbilt University. His research is focused on enhancing nanotechnology for optical components of solar panels. His insight would be valuable for clarifying many technological barriers we may encounter.

3. Abhishek Dubey
Prof. & Researcher at Vanderbilt University
Contact: abhishek.dubey@vanderbilt.edu

Dr. Dubey is a researcher and professor at Vanderbilt University. He is part of the team focused on smart cities and focuses on cyber-physical systems. His insight would be helpful for explaining how the integration of solar roadways may occur.

2. Scott & Julie Brusaw
Founders of Solar Roadways
Contact: Presentations@SolarRoadways.com

These are the originals founders of the solar roadways concept. They are based in Idaho but are well known to respond to student researchers. They will have valuable insight all around for the concept.

Other potential contacts include Nashville Metro Department of Transportation, although finding specific sustainability contacts is somewhat difficult. The Tennessee Department of Transportation could also be a point of contact.

Interview summaries

Placed in the Final Report Appendix.

Revised Section 1-5 analysis

Incorporated technology intelligence into Sections 1-5.

Contact log

Who contacted	Contact	Date	Success?	Referrals	Follow-up Actions
Juliette	Scott & Julie Brusaw	4/4	Yes	...	...
Pruett	Thomas Rush	4/5	Yes	...	Sent follow-up on 4/6
Sophie	Dr. Caldwell	4/5	Yes	...	

Summary of Strengths, Weaknesses Opportunities, and Threats (SWOTs Analysis)

Summary of Strengths, Weaknesses Opportunities, and Threats

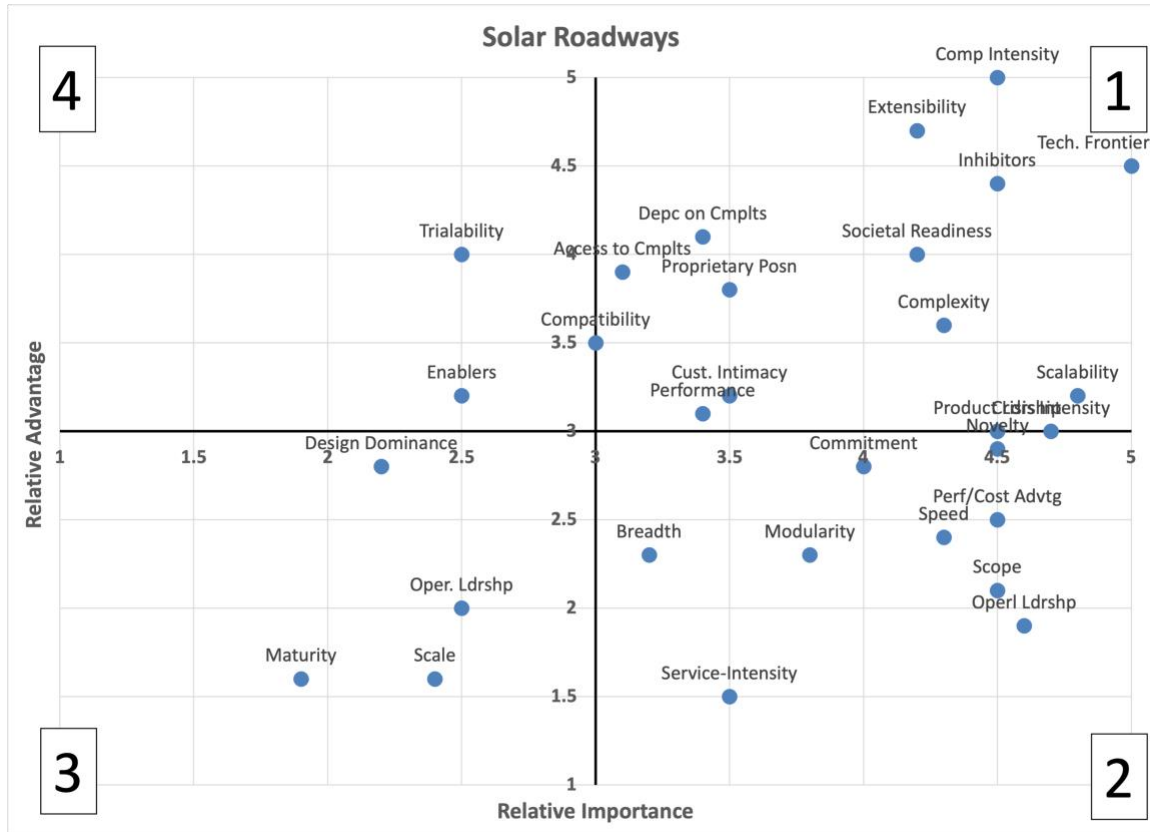


Figure 16: SWOT Analysis of ISF's.

Strengths:

1. Position Relative to the Technological Frontier

Position relative to the technological frontier measures how advanced a technology is compared to the best performance currently possible in its field. This factor evaluates whether a product represents a meaningful technological step forward rather than a small improvement to existing solutions. Solar roadway technology performs relatively well on this factor, as seen in Figure 16, because it combines multiple systems that have traditionally been separate. Instead of building roads and solar panels independently, the technology integrates electricity generation directly into transportation infrastructure through load bearing photovoltaic panels. Companies such as Colas have already developed and tested solar pavement products like Wattway, which demonstrates that the concept has moved beyond theory and into real world experimentation (Colas,

Wattway by Colas, 2023). These early projects show that the technology is expanding the capabilities of traditional infrastructure by integrating renewable energy systems directly into transportation networks.

Since solar roadways score well on position relative to the technological frontier, this factor creates an important long-term opportunity for the technology. Public awareness of climate change has grown significantly in recent years, which has led governments and communities to place greater importance on renewable energy development and reducing carbon emissions. Infrastructure that contributes to clean energy production is becoming increasingly valuable as governments work toward long-term sustainability goals. Since solar roadways represent a creative form of renewable infrastructure, they may attract greater attention from investors, receiving more funding and support. In addition to producing clean energy, solar roadway systems can include embedded heating elements that help melt snow and ice during winter conditions. This capability could improve roadway safety and reduce reliance on traditional snow removal methods. It could also decrease the need for chemical de-icing treatments that require significant public spending and can harm surrounding ecosystems. For companies developing solar roadway technology, the most important strategic step is continuing to improve the system through testing and demonstration projects. This will help ensure the technology can meet the performance standards required for large-scale infrastructure use (International Energy Agency, 2020). To capitalize on this technological advantage, developers should focus on pilot projects in regions with high winter maintenance costs, such as Minnesota or Michigan. Short solar pavement segments near municipal buildings or university campuses in these locations could test the effectiveness of built-in heating systems for melting snow and ice. These targeted demonstrations would provide measurable data on safety and cost savings while helping attract additional infrastructure investment.

2. Extensibility

Another one of Sunstrider Pavement's main strengths is extensibility. Traditional pavement and standalone solar panels were never designed to do more than their singular purpose. Asphalt supports vehicles and solar panels generate electricity. Neither was meant to or can do both. Our photovoltaic pavement fundamentally redefines what a paved surface can be. Extensibility is one of the most heavily weighted innovation success factors in our analysis because modern infrastructure is becoming increasingly more multipurpose. Unlike solar canopies or roadside panels, which are largely limited to electricity generation, our pavement system can integrate snow and ice mitigation, LED lighting for traffic signaling and wayfinding, and additional smart city technologies such as sensors or data collection modules. In dense pedestrian and campus environments

where space and budgets are constrained, the ability to stack multiple functions into the same physical footprint is a massive competitive advantage. Where extensibility is a limitation for incumbent approaches, it is a defining strength for us.

Extensibility is defined as the ability of the technology to support additional complementary functions beyond core electricity generation. As shown in Figure 17, photovoltaic pavement scores highest on this factor compared to the incumbent technologies. Municipalities, campuses, and private developers are increasingly prioritizing multi-functional infrastructure that enhances safety and enables smart-city integration without requiring additional land use. Solar canopies and roadside panels cannot easily evolve beyond power generation without separate installations and additional spending. In contrast, our pavement platform is designed to perform multiple functions in one system. Strategically, this positions us not merely as an energy provider, but as an extensible infrastructure platform. By marketing the pavement as a long-term, upgradeable system rather than a static installation, we capitalize on our strength in extensibility and align directly with the market's demand for adaptive, intelligent infrastructure. Choosing a host company which already has a good reputation in this market is essential to the survival of our product.

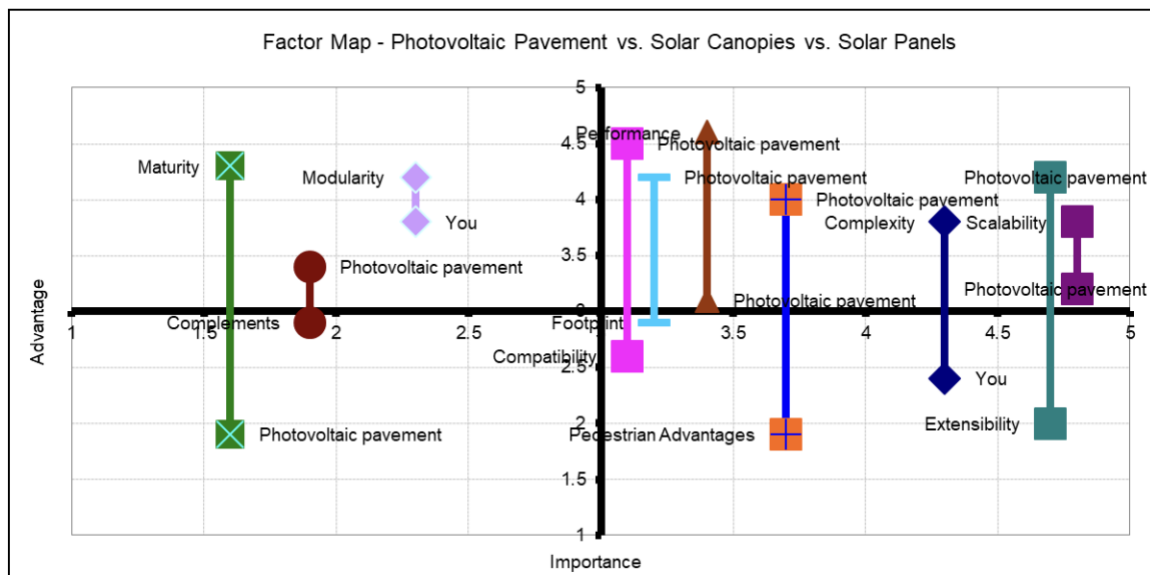


Figure 17: Importance-Advantage Chart for Photovoltaic Pavement vs. Solar Canopies vs. Solar Panels

Weaknesses:

1. Technological Complexity

Technological complexity refers to how difficult a technology is to develop and operate relative to existing solutions. Solar roadways are considered highly complex

because they combine several functions into a single piece of infrastructure. The road surface must be strong enough to support vehicles and withstand weather conditions while also containing photovoltaic cells and electrical components that generate power. For example, Colas explains that its Wattway solar road panels contain photovoltaic cells protected by several layers of materials that allow sunlight to pass through while protecting the system from vehicle traffic and weather damage (Colas, 2015). Since these panels must function both as a durable road surface and as an energy-producing system, the engineering requirements are more demanding than those of traditional asphalt or concrete roads. Researchers have also found that reduced sunlight exposure from flat solar panels and shading from vehicles can lower the efficiency of energy production from solar pavement systems (Lv, Zha, Hu, Lei, & Niu, 2025). Therefore, technological complexity remains a weakness for solar roadways since it increases costs and creates additional challenges for broad implementation.

Technological complexity creates important long-term risks for solar roadway development because infrastructure systems are expected to operate reliably for decades with minimal disruption. Transportation agencies and infrastructure investors tend to favor technologies that are simple to maintain and predictable in performance. If solar roadway systems remain highly complex, maintenance could become more expensive and time consuming than traditional pavement systems, which may discourage governments from adopting the technology at scale. Research on solar pavements also suggests that maintaining long-term durability and energy efficiency in real roadway environments remains a challenge, particularly when systems are exposed to continuous traffic and environmental stress (Hu, Vizzari, Zha, & Mantalovas, 2023). If early projects fail to meet reliability expectations, public confidence in the technology could decline and slow further investment. Due to this risk, companies developing solar roadway systems will need to prioritize long-term durability testing and design improvements so the technology can meet the reliability standards expected for transportation infrastructure. To manage technological complexity, developers should first deploy solar pavement in lower-traffic environments such as bike paths and pedestrian walkways. For example, retail or stadium parking lots could use solar pavement to generate electricity for lighting or nearby electric vehicle charging stations while allowing engineers to test durability before highway deployment.

2. Service Intensity

Sunstrider faces another real weakness in service intensity. Service intensity is defined as the level of technical support, maintenance, and operational expertise required after installation. Unlike conventional asphalt, which public works departments already maintain regularly, photovoltaic pavement integrates structural materials with electrical generation, heating systems, and other monitoring. This complexity increases the chance

that any failures will result in long repair times and difficult service. Trained technicians will be required to service photovoltaic pavement. At the same time, conventional photovoltaic systems remain significantly more efficient and less expensive on a per-watt basis than embedded solar pavement (National Renewable Energy Lab, 2024). Because customers can have reliable energy production through traditional pavement paired with roadside solar panels, they may hesitate to adopt a system that introduces additional maintenance complexity. Even if modular panels and remote monitoring reduce downtime, the perception of higher service demands and vendor dependence remains a barrier to adoption.

To account for this weakness, Sunstrider must be very particular with respect to its commercialization strategy. The ideal market for the early stages of this product would be campuses and sidewalks of private developments. These areas have a green incentive to install our product and would be willing to sacrifice the potentially higher maintenance cost to improve the user experience. Sidewalks are also easier to troubleshoot than paved roads where shutdowns are much more costly. Our strategy should also emphasize long warranties, reliability guarantees, and demonstrate the ease of repair to outside contractors. Once again, choosing a host company with a strong reputation in reliability is key to exploiting this weakness. The host company should also have a large employee base with many technicians, ready to service many of our products.

3. Compatibility

Compatibility refers to how well a new technology integrates with existing infrastructure systems, construction practices, and operational processes. For infrastructure technologies such as roadways, compatibility is particularly important because transportation agencies and contractors rely on standardized construction materials and procedures that have been proved through decades of real-world performance and testing. Solar roadways currently face compatibility challenges because they introduce a fundamentally different pavement structure compared to traditional asphalt or concrete roads. Conventional roadway systems are built using materials and installation processes that are widely understood by construction contractors and public infrastructure agencies. Solar roadway panels, however, integrate photovoltaic cells, electrical wiring, heating elements, and lighting systems within the pavement surface. This would require more training for install and maintenance. These additional components create technical and operational differences that may complicate integration with existing road construction and maintenance practices (Federal Highway Administration, 2020).

These compatibility challenges have important strategic implications for the development and marketability of solar roadway technology. Transportation agencies typically prefer infrastructure solutions that can be implemented within existing construction standards and maintenance procedures. If solar roadways require entirely new installation methods or specialized maintenance expertise, adoption may be slowed by institutional resistance and regulatory uncertainty. One potential strategy to improve compatibility is to design solar roadway panels as modular units that can be installed and

replaced using processes similar to conventional pavement maintenance. In addition, early deployment should focus on environments such as campuses, bike paths, and pedestrian walkways where infrastructure requirements are less rigid than those governing major highways. There will also be less traffic and wear on the product. These environments provide opportunities to test the technology while gradually improving compatibility with existing infrastructure systems (Shekhar, et al., 2018).

Opportunities:

1. Societal Readiness

Societal readiness refers to how willing governments and the general public are to accept and support a new technology. For solar roadways, this factor represents a clear opportunity because public interest in renewable energy has increased as concerns about climate change and energy security continue to grow. Recent research shows that about two-thirds of Americans support transitioning the economy toward clean energy sources in the coming decades, and many believe developing renewable energy should be a national priority (Fernandez, 2025). This social support creates a favorable environment for new renewable technologies because communities and governments are more open to testing infrastructure that can contribute to cleaner energy systems. Solar roadways benefit from this trend because they offer a visible way to combine transportation infrastructure with renewable power generation. As a result, strong public interest in clean energy can make it easier for companies to secure partnerships and early investment needed to continue developing the technology. As development continues, companies working on solar roadway systems will need to focus on public demonstration projects and partnerships with local governments to take advantage of growing support for renewable infrastructure.

Societal readiness also has important strategic implications for solar roadway development because public attitudes toward climate change and renewable energy influence how willing communities and governments are to support new infrastructure ideas. When people recognize climate change as a serious issue, they are generally more open to solutions that help reduce emissions and expand clean energy. Recent polling shows that about 63 percent of Americans believe the effects of global warming are already happening, which reflects a growing awareness of climate related risks and the need for action (Saad, 2026). This level of concern creates an opportunity for technologies like solar roadways because policymakers may feel greater pressure to invest in visible sustainability projects that demonstrate progress toward climate goals. At the same time, this social momentum raises expectations for performance. If early projects show clear benefits, the technology could gain credibility and attract additional investment. If they fail to perform reliably, public trust could weaken and slow future

adoption. Therefore, companies developing solar roadway systems must use this favorable social climate carefully by prioritizing well-designed pilot projects that build long-term confidence in the technology. Developers can leverage public support for renewable energy by partnering with cities that prioritize sustainability initiatives, such as San Diego or Austin. Solar powered pedestrian walkways or bike lanes in parks and urban greenways would allow the public to see the technology in action while generating electricity for nearby lighting or public charging stations.

2. Scalability

Scalability is a major opportunity for photovoltaic pavement as infrastructure modernization and renewable energy usage are growing rapidly. The U.S. highway, street, and bridge construction industry generates over \$190 billion annually, offering an infrastructure market that could support long-term success if adoption barriers are addressed (Pigott, 2025). At the same time, solar energy continues to account for a significant share of U.S. renewable energy sources. This shows that both the private and public sectors are ready for wider adoption (U.S. Energy Information Administration, 2025). These trends create strong conditions for our product, but scaling solar roadway technology requires more than just market demand. It depends on reliable supply chains, regulatory approvals, and integration into established construction workflows. This is different from rooftop solar, which benefits from large installer networks and recent cost declines (National Renewable Energy Lab, 2024). Solar pavement must scale within the more conservative and regulation-heavy civil infrastructure sector.

To properly scale our product, our strategy should focus on scaling slowly at first to prove our technology. As mentioned before, smaller projects on campuses and private developments would bolster Sunstrider's reputation before continuing to larger markets like highway infrastructure. The trial phase for this may be quite long, potentially many years, before we see any benefit, since this technology must prove its long-term reliability. Over time, as manufacturing volume increases and design standardization reduces per-unit costs, the product can move up-market into larger municipal and transportation projects. In this case, it's key to have a host company with a large amount of capital who is willing to take a loss in the short-term. They must believe in the products mission and lobby for green tax cuts. All of this requires a large corporation.

Threats:

1. Competitive Intensity

Competitive intensity refers to how much competition a new technology faces from solutions that are already widely used. For solar roadways, this creates a major threat because existing technologies already perform the same roles more efficiently. Traditional solar panels installed on rooftops or in large solar farms produce electricity more effectively because they can be placed at angles that capture more sunlight and are not affected by traffic conditions. Solar panels embedded in road surfaces remain flat and

can receive less sunlight when vehicles pass over them or when dirt collects on the surface, which can reduce the amount of energy they generate. Traditional asphalt roads are also much cheaper to build and maintain than solar pavement systems. Research comparing solar pavement with standard roads shows that solar roadways require more materials and specialized components, which can increase costs and make maintenance more difficult over time (Hu, Vizzari, Zha, & Mantalovas, 2023). Since these established technologies are already familiar to infrastructure planners and the public, decision makers may prefer traditional roads combined with separate solar installations rather than adopting solar roadway systems.

Competitive intensity also creates long-term strategic challenges for solar roadway technology because it must compete directly with industries that are already highly developed and continuously improving. Traditional solar energy technology continues to become more efficient and less expensive each year as manufacturing improves and global production increases. According to the International Renewable Energy Agency, the cost of electricity generated from solar photovoltaic systems has dropped dramatically over the past decade as the technology has matured and scaled globally (International Renewable Energy Agency, 2025). This means that solar roadways are not only competing with current solar installations but also with a technology that is rapidly becoming more affordable and efficient over time. At the same time, the road construction industry has decades of experience refining asphalt and concrete pavement systems that are reliable and relatively inexpensive to maintain. Since these competing industries continue to improve their technologies, solar roadways face the strategic risk of falling further behind if development does not keep pace with advancements in both renewable energy and traditional infrastructure. As a result, the long-term success of solar roadway systems will depend on whether developers can demonstrate advantages that competing technologies cannot easily replicate. Due to this competitive pressure, solar roadway developers must identify niche applications where the technology provides unique value before attempting broader infrastructure deployment. Promising locations include airport service roads and remote highways where electricity from the panels could power lighting or EV charging infrastructure. These environments allow solar roadways to provide benefits that traditional roads or separate solar installations cannot easily replicate.

2. Trialability

Another main threat to our product is trialability. Trialability refers to the extent to which a technology can be tested on a limited scale before committing to full release. In theory, solar roadways are well suited for small pilot installations in sidewalks, plazas, bike paths, or parking stalls. This allows municipalities, universities, and corporate campuses to evaluate durability, safety, and energy performance without exposing high-

traffic roadways to risk. These contained environments also align with sustainability initiatives. However, trialability also presents a strategic threat. Even small pilot installations can carry significantly higher upfront costs than conventional pavement alternatives, which may discourage experimentation or lead to highly scrutinized public demonstrations (Anthony, 2016). Additionally, because pilot projects are often highly visible, any technical shortcomings (such as panel failures, lower-than-expected energy output, or maintenance challenges) can disproportionately shape public perception. It's a huge risk to the publicity of our product.

To reduce this threat, any public installations of the early-stage versions of our product must be carefully selected. There should be minimal public media coverage and signage, unless necessary for regulations or safety. While we can't place it in a low foot traffic area (that would give biased results), we should select an area where users are used to seeing innovation. A campus sidewalk would be a perfect test ground for the first installation. College campuses are no stranger to weird inventions and media coverage would be limited to just the student body. If something breaks, it's viewed as more of a learning opportunity than a failure. Additionally, we could incentivize student researchers into the pavement once their success is proven. This means our host company needs to have a strong connection to university contractors or be nearby to a local college that is open to innovation.

3. Technology Maturity

Another one of the primary threats identified in the matrix is technology maturity, which appears in the low importance and low relative advantage quadrant. Although this factor may not currently be the most critical concern for the market, it still represents a potential challenge. Traditional road infrastructure has been developed and refined over many decades. Asphalt and concrete technologies are well understood, widely available, and supported by established construction industries. Solar roadway systems face several engineering challenges, including reduced efficiency compared with conventional solar installations and durability concerns related to heavy traffic and environmental exposure. The lack of long-term performance data may create uncertainty for potential adopters, particularly for governments responsible for maintaining critical infrastructure.

To overcome this threat, Solar Roadways should focus on collecting long-term performance data from pilot installations. Demonstrating durability, reliability, and maintenance requirements will be essential for building confidence among infrastructure planners and transportation agencies. Continued research and development will also help improve the technology and move it closer to the level of maturity seen in traditional road construction methods.

Intellectual Property Strategy

Patent strategy

Since seeking a patent is costly and time intensive, the decision to pursue a patent for Solar Roadways should rest on whether the invention clearly satisfies the legal requirements for a utility patent. In the United States, an invention must be novel, nonobvious, and useful to qualify for protection (U.S. Patent and Trademark Office, Nonprovisional (Utility) Patent Application Filing Guide, 2023). The central patentable element of Solar Roadways is not the use of solar panels in transportation infrastructure, since photovoltaic systems are already widely deployed in other settings (U.S. Department of Energy, n.d.). Instead, the innovation lies in the modular roadway panel design that integrates load bearing textured glass, embedded photovoltaic cells, programmable LED lighting, heating elements, and internal electronics into a surface engineered to withstand vehicular traffic. This integrated system transforms a conventional road into an energy-generating platform.

Solar Roadways appears to meet the novelty requirement because no single prior design combines structural roadway functionality with embedded solar generation, lighting, and heating within one unified panel architecture (U.S. Patent and Trademark Office, U.S. Patent and Trademark Office, 2026). It also presents a credible argument for nonobviousness. Road surfaces must endure significant mechanical stress, temperature variation, and environmental exposure, and federal transportation research emphasizes how demanding these conditions are for pavement materials (Federal Highway Administration, 2020). Integrating photovoltaic technology into a surface capable of meeting these standards involves more than substituting materials and requires coordinated engineering. Finally, the invention clearly satisfies the usefulness requirement because it produces renewable electricity and reduces ice accumulation through built-in heating (U.S. Patent and Trademark Office, Nonprovisional (Utility) Patent Application Filing Guide, 2023; Federal Highway Administration, 2020). Given these factors, Solar Roadways should pursue a utility patent because it would be a strategically sound step to protect the substantial research and development investment necessary to commercialize this technology.

In addition, our chosen host company Colas already has an approved U.S. patent for photovoltaic roadway structures (United States of America Patent No. US11075311B2, 2014). Their patent protects a roadway structure with a photovoltaic cell embedded under the outer surface, which has some sort of texture for grip from vehicles. We would likely file a separate patent based off their technology to include our other features for pedestrian safety, but this is fantastic starting position for our host company.

Other strategies for building your proprietary position

Instead of relying only on patents, Solar Roadways can build a strong proprietary position through system integration, data control, and long-term institutional partnerships. Because Solar Roadways is a complex system that combines energy generation, roadway infrastructure, sensors, and digital monitoring, it is difficult for competitors to copy the

full platform rather than just individual parts. This creates a natural barrier to entry based on system complexity rather than legal protection alone.

Another key strategy is data ownership and learning advantages. As Solar Roadways systems are deployed, they generate data on traffic flow, energy production, weather conditions, and system performance. Over time, this data can be used to improve efficiency, maintenance planning, and system design, allowing Solar Roadways to improve faster than competitors and build an experience-based advantage.

Solar Roadways can also strengthen its position through long-term relationships with public institutions, such as cities, universities, and transportation agencies. By becoming part of infrastructure planning and pilot programs, the company can create high switching costs and become embedded in future development plans, making it harder for competitors to replace them.

Finally, service-based integration, such as maintenance contracts, monitoring systems, and energy management services, can shift Solar Roadways from being a product supplier to an infrastructure partner. This creates long-term dependence on the system and strengthens its proprietary position without relying only on patents.

Product Strategy

Product-market scope

Market development represents the most appropriate product-market scope for Solar Roadways at its current stage. Expansion should focus on adjacent environments such as sidewalks, bike paths, and airports rather than immediate deployment on highways. This direction aligns with the multifunctional nature of the technology. Solar Roadways panels combine energy generation with embedded lighting and control systems, which means the value of proposition extends beyond electricity into safety, and infrastructure intelligence (Solar Roadways, The Solar Road Panel, n.d.). These benefits are easiest to see in controlled settings where performance can be directly observed and safety improvements are clearer than in high speed road environments (Solar Roadways, The Solar Road Panel, n.d.).

Market penetration into traditional road infrastructure introduces substantial risk and should not be prioritized in the near term. Although highways represent the original long term vision, the technology has not yet demonstrated the reliability or economic feasibility required for such conditions. Existing pilot installations have primarily functioned as testing grounds, revealing technical challenges that remain under refinement (Solar Roadways, The Solar Road Panel, n.d.). Engineering evaluations reinforce this limitation by noting that testing has largely occurred in parking lot settings, which are less demanding than highways due to lower speeds and reduced structural stress (Ronald A. Coutu, Newman, Munna, Tschida, & Brusaw, 2020). Broader academic

research similarly highlights ongoing concerns with durability and cost effectiveness in photovoltaic pavement, particularly under heavy traffic conditions (Dai, Yin, & Lu, 2021). Expanding further into the core roadway market at this stage would expose the company to risks that are not yet manageable given the current maturity of the technology.

Product development is a useful second step because it makes the overall system stronger and more practical. Adding features such as microgrid controls and simple management software would make the technology more reliable and easier to use. The U.S. Department of Energy notes that solar systems with storage can keep running during power outages, which increases their value (U.S. Department of Energy, n.d.). Solar Roadways already includes features like built in electronics and cable channels, so expanding into these areas fits naturally with what has already been developed (Solar Roadways, The Solar Road Panel, n.d.). However, these improvements only matter if there are clear markets where the full system can actually be used.

Diversification is not advisable at this stage because it introduces both technological and market uncertainty. The core product still requires continued validation and improvement, and pursuing entirely new applications would divide resources and increase execution risk. Existing research continues to identify unresolved technical challenges, particularly related to long term durability and performance under load, which suggests that focus should remain on refining the current technology rather than expanding into unrelated areas (Dai, Yin, & Lu, 2021).

Market development offers a practical path forward by taking advantage of where the technology already fits well. Applications such as parking areas, bike paths, and pedestrian spaces show clear value and align with how the product is currently designed (Solar Roadways, The Solar Road Panel, n.d.). Public sector research also supports expanding into these types of spaces, noting strong potential for solar integration in transportation related properties and the ability to reduce adoption barriers through flexible financing (Federal Highway Administration, 2020). Work from the California Department of Transportation further highlights solar powered bike paths as a promising example of dual use infrastructure, especially when introduced through smaller pilot projects (Lee, Mazarei Saadabadi, & Martinez-Morales, 2025).

This approach allows Solar Roadways to build credibility and improve performance. As the technology develops, success in these areas can support a gradual move into more demanding roadway applications.

Product family

The introductory product in the Solar Roadways product family will be modular solar pavement systems designed for pedestrian environments, such as sidewalks, bike paths, campus walkways, and parking lots. This entry product would be the most viable entry option as it would match the current state of technology and avoid exposure to high-speed vehicle risks. The target audience would include universities, corporate campuses, local governments, and development organizations interested in prominent

sustainability initiatives and smart infrastructure solutions. These areas would offer the benefits of integrated solutions such as solar panels, LED lights, and programmable road surfaces, making the product viable for early adoption.

The next logical step in this basic product is a “smart pavement system package,” which combines energy storage, grid connection, and panel control software. This is a direct extension of the panel’s capabilities and adds to the value proposition of the basic product. This is especially useful for campuses and smart cities. For instance, the use of solar pavement and storage ensures that infrastructure is always available during outages, providing a safe and resilient infrastructure. This is a “system” product that is sold to institutional customers and can be delivered with a software package for monitoring energy production, health monitoring, and lighting control. There is increasing public-sector support for the integration of energy systems into transportation infrastructure, and there is recognition of the potential for “leverage of transportation corridors for distributed renewable energy generation” (U.S. Department of Transportation, 2022).

The next product in the product family would be a transport adjacent infrastructure product, expanding on applications such as taxiways at airports, service roads, and low-speed logistics areas. This would provide a higher load requirement but would still be more controlled than a highway environment, providing a natural progression from the previous product. The target market would include airports, industrial parks, and logistics areas. However, as mentioned in the research, durability and structural performance are still major issues, as solar pavement systems need to provide transparency, strength, and resistance to wear and tear under load (Cai, 2025). This product would therefore need to collaborate with established infrastructure and construction companies.

In order to encourage more users and reduce entry costs, a derivative product of the product family would be a retrofit or add-on system, which comprises smaller modular panels or kits that can be retrofitted into existing pavement areas (e.g., parking stalls, crosswalks, or bike lane segments). This product would target municipalities and property owners interested in testing the product at a lower cost and risk. This is especially relevant since solar roadway systems currently are significantly more expensive than traditional pavement systems, costing several times more per mile and per unit of energy produced. Similar to the add-on display product, this product would also include optional installation services through certified contractors, making it accessible to users without specific skill sets.

Finally, a longer-term product in the family is full roadway integration for vehicular traffic, representing the original vision of solar roadways. However, this remains a future-stage product that depends on improvements in durability, efficiency, and cost. Existing pilot projects and analyses highlight that solar roadways currently struggle with lower energy efficiency due to flat orientation and shading, as well as challenges in handling heavy traffic loads (Marquart, 2021). Entry into this segment would therefore require partnerships with state DOTs, major construction firms, and federal agencies, along with continued advancements in materials and panel design.

Product specifications

For Solar Roadways to meet user needs and function effectively in real world environments, the product must satisfy a small set of clear and measurable requirements.

These focus on safety, durability, and basic functionality in its initial form, while also outlining performance goals as the technology matures.

Solar Roadways shall:

- I. Support vehicle loads of at least 80,000 pounds without structural failure, matching standard roadway weight requirements
- II. Provide a surface traction coefficient equal to or greater than conventional asphalt under both dry and wet conditions
- III. Generate usable electrical output under typical daylight conditions with no more than a 20% performance drop due to surface wear or environmental exposure

Solar Roadways should:

- IV. Maintain functional performance for at least 20 years with limited degradation in strength and energy output
- V. Allow individual panels to be replaced within a short service window, ideally under one hour per panel
- VI. Deliver a total lifecycle cost that is competitive with traditional paving when energy generation and maintenance savings are included

Potential lead users and minimum viable product

Identify a lead user in your target market and an initial (minimum viable) product. What 3-5 key pieces of information will you capture from this lead user? How would you capture it?

A strong lead user for Solar Roadways would be a large university campus, such as those with established sustainability initiatives and significant pedestrian infrastructure. Universities represent an ideal testing environment because they have high foot traffic, controlled vehicle exposure, and strong incentives to adopt visible, innovative sustainability solutions. Additionally, campuses often have access to funding for pilot projects and are willing to experiment with emerging technologies to enhance both safety and environmental performance. These characteristics make them highly motivated to adopt and refine solar pavement systems while facing relatively low risk compared to highway agencies.

The minimum viable product (MVP) would consist of a small-scale installation of modular solar pavement panels deployed in a defined pedestrian zone, such as a campus walkway, bike path, or parking area. This MVP would include core functionality: energy generation, embedded LED lighting for visibility and signaling, and basic durability sufficient for pedestrian and light vehicle loads. The goal of the MVP is not to fully optimize performance, but to demonstrate feasibility, provide immediate value through

lighting and energy generation, and allow real-world testing of installation, maintenance, and user interaction.

From this lead user, we can collect several key pieces of information. First, performance data related to energy generation, durability, and maintenance frequency is critical to understanding whether the system meets operational expectations. Second, user experience feedback should be collected, particularly regarding perceived safety, visibility, and overall usability of the surface. Third, installation and lifecycle cost data should be tracked to evaluate economic feasibility and identify cost drivers. Fourth, different operational insights such as ease of panel replacement and system reliability under daily traffic should be documented. Finally, stakeholder perception, including how students, faculty, and administrators value the system, and if they would implement it on a larger scale, will be assessed to understand its potential for broader adoption.

This information can be captured through a combination of embedded sensors within the panels, maintenance logs, and energy monitoring systems, alongside surveys and interviews with campus-goers. These data sources will provide both quantitative and qualitative insights, enabling iterative improvement of the product and supporting a clearer path toward broader market adoption.

Product-market penetration path

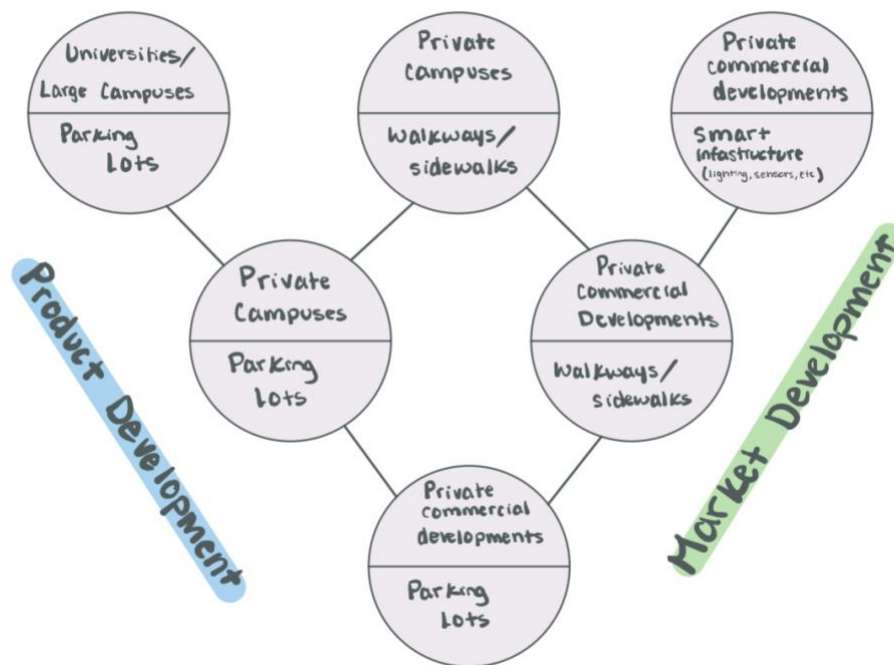


Figure 18: Bowling Pin Model for Solar Roadways

The bowling pin model begins with a focused beachhead market of private commercial developments, specifically targeting parking lots as the initial application. This segment is ideal because it has low traffic demands, fewer regulatory barriers, and

clear value through energy generation and sustainability branding. After establishing success in this niche, the strategy expands in two directions: market development, by applying the same parking lot solution to similar environments such as campuses and universities, and product development, by introducing new applications like walkways and smart infrastructure within the same customer segment. By leveraging proven performance and customer references at each step, the technology can gradually move into more complex and larger-scale markets, following the step-by-step “bowling pin” approach.

Operational Strategy

Operational Architecture of the Business

The operational architecture of Sunstrider Pavement focuses on two primary activities: 1) developing durable, multifunctional solar roadway panels and 2) manufacturing and deploying these systems at scale within targeted environments such as sidewalks, campuses, and controlled roadway settings. As shown in the high-level SIPOC framework (Figure 19), the business relies on a network of suppliers including photovoltaic component manufacturers, materials providers, acquiring engineering talent, capital investors, and logistics partners. These suppliers provide key inputs such as solar cells, composite materials, electronics, manufacturing facilities, and financial resources. Core processes include development, increasing the value of the product, and customer marketing. This includes research and development of load-bearing photovoltaic panels, system integration (lighting, heating, and sensing technologies), manufacturing and assembly, and installation and maintenance. The outputs of these processes are modular solar roadway panels ***which result in smart system integration. These outputs are delivered to customers such as municipalities, transportation departments, commercial developers, and institutional campuses. For Sunstrider to be viable, the operational architecture must meet critical requirements including scalability, cost competitiveness relative to traditional pavement and solar alternatives, long-term durability under traffic loads, and the ability to deliver differentiated value through multifunctionality and smart infrastructure capabilities.***

Suppliers	Inputs	Processes	Outputs	Customers	Requirements
Photovoltaic manufacturers	- Solar cells	- R&D of solar pavement panels - IP for covered cells - Hire engineers	- Modular solar panels - Patent	- Municipal governments - Private developers - Campuses	- High durability (traffic loads) - Simple replacement - Efficiency comparable to normal solar cell
Composite materials suppliers	- Tempered glass - Roadway composites (like concrete or rubber)	- Utilize previously developed roadway tech	- Non-opaque composite roadway material	- Departments of Transportation	- Cost competitiveness - Easy manufacturing
Electronics suppliers	- LEDs - Sensors - Wiring systems	- Integrate smart features - Work with software teams	- Smart-enabled roadway panels - Smart city possibilities	- Smart city planners - Municipalities	- Reliability - Weather resistance - Energy efficiency
Software engineers	- Electronic systems - Control software - Automation for cities - Operators	- Use prototype to develop - Build in safety measures	- Smart city software control - Potential data sources	- Data analysts - City planners - Traffic controllers - Public works	- Robust safety measures - Trained operators - Troubleshooting team
Labor pool	- Engineers - Plant operators - Installation teams	- Hiring and training - System design and testing	- Skilled workforce - Optimized system designs	- Internal operations	- Talent quality - Operational efficiency
Investors / Government	- Capital - Grants	- Fund R&D and scaling	- Scaled production capabilities	- Internal growth	- ROI - Sustainable scaling
Infrastructure & logistics	- Manufacturing plants - Transport networks	- Manufacturing - Distribution - Installation	- Installed solar roadway systems	- Contractors - Developers	- Scalability - Timely delivery - Installation efficiency

Figure 19: High-level SIPOC diagram highlighting key components of the operational strategy

Key Processes

Three key processes are selected from the high-level SIPOC diagram for detailed development: R&D and panel development, manufacturing and assembly, and installation and maintenance. These processes were chosen because they represent the highest-risk and highest-value activities in Sunstrider's operational architecture. If any one of them fails to meet performance requirements, the entire commercialization

pathway stalls. The SIPOC diagrams for each process are presented below in Figure 20, Figure 21, and Figure 22.

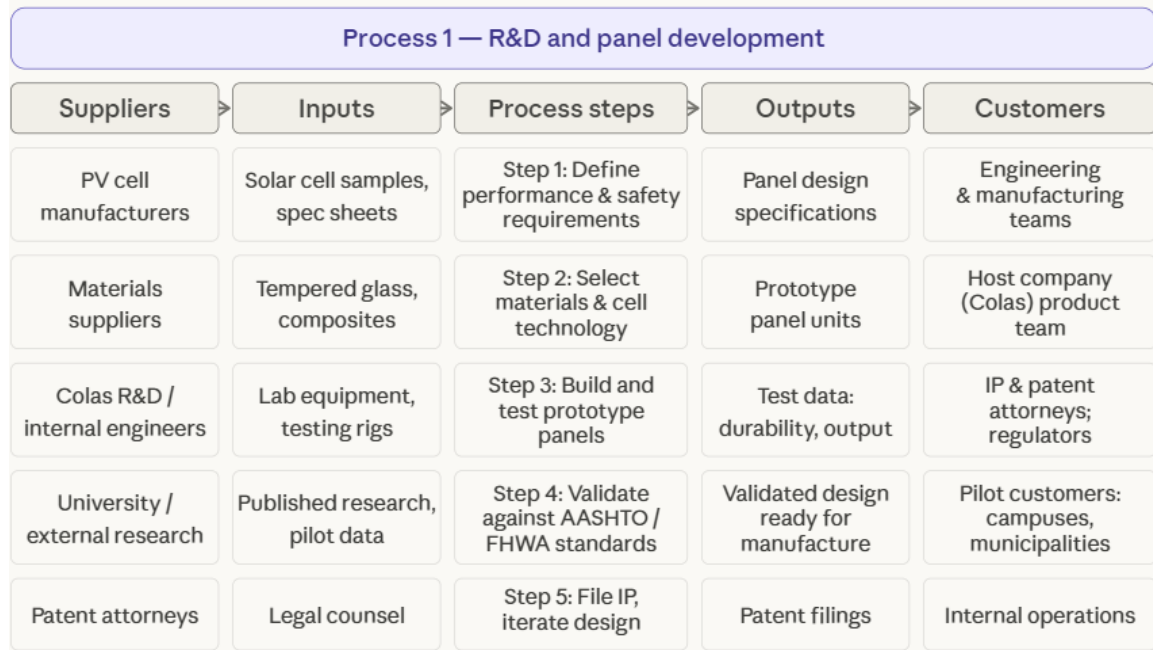


Figure 20: R&D and Panel Design SIPOC

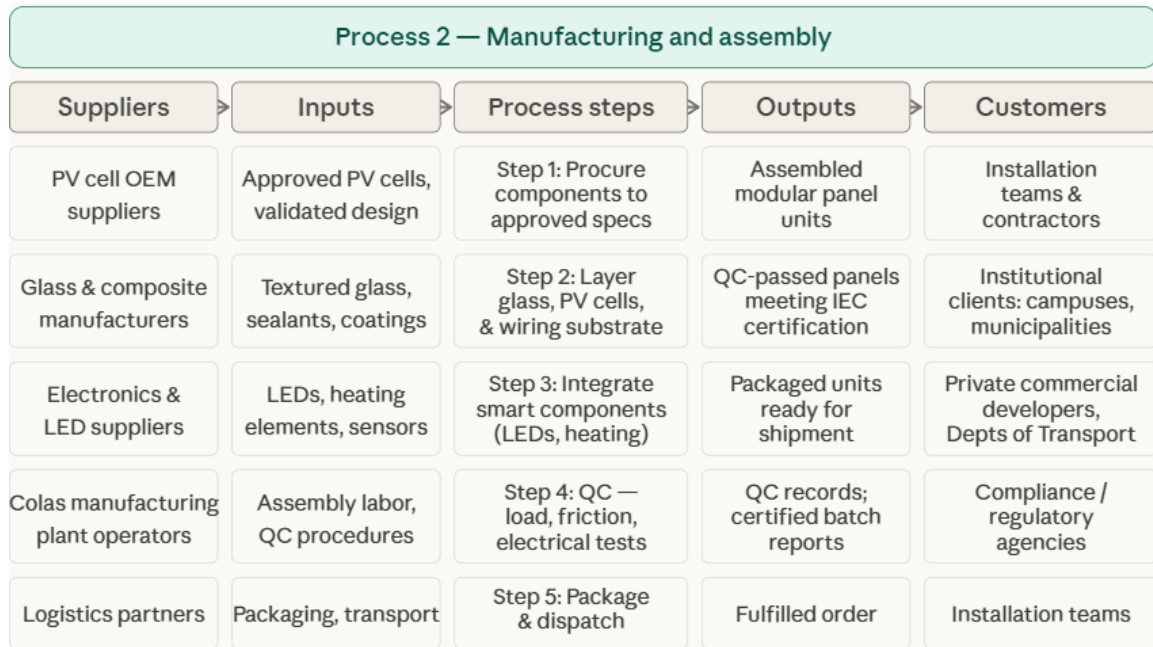


Figure 21: Manufacturing SIPOC

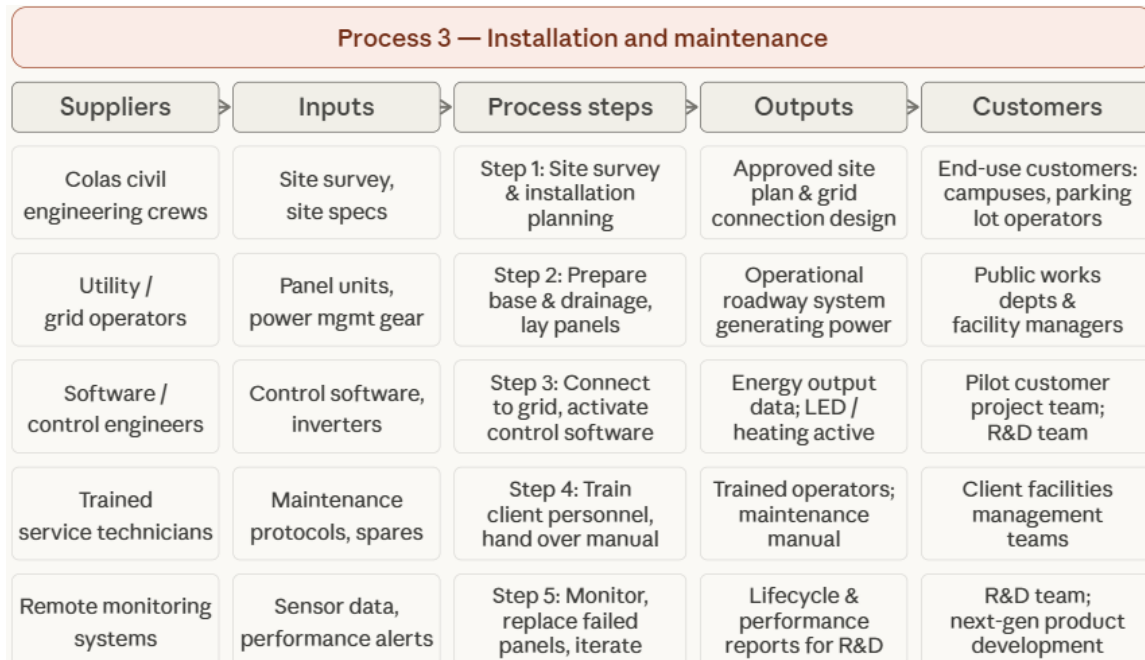


Figure 22: Installation and Maintenance SIPOC

Sourcing

With Colas as the host company, the operational approach should prioritize building internal capabilities around the elements of the solar roadway system that are most technically complex and strategically valuable. Given Colas's existing strengths in transportation infrastructure, materials engineering, and large-scale project execution, it is well positioned to take ownership of the design and production of the solar roadway panels. These panels represent the core of the innovation, integrating photovoltaic generation and embedded technologies. Keeping these processes internal allows Colas to maintain strong oversight of performance while steadily refining the technology and safeguarding its proprietary designs as the system evolves.

In addition to manufacturing, system integration should remain an internal focus. The functionality of solar roadways depends on the physical panels and the coordination between energy generation and connectivity to external systems. Developing these capabilities within the firm allows Colas to establish a cohesive platform rather than a standalone product. This approach strengthens its ability to differentiate in the market and positions the company to capture value from both infrastructure and technology-driven services over time.

By contrast, activities that are standardized or already performed efficiently by specialized providers are more appropriate for outsourcing. Key inputs such as photovoltaic cells and electronic components can be sourced from established manufacturers that benefit from scale and cost efficiencies. Similarly, logistics and certain aspects of installation can be contracted to third-party providers, particularly in regions where local expertise is advantageous. Collaboration with utility companies for grid integration is also essential, as these entities manage existing energy infrastructure and regulatory requirements.

Over time, several internally developed capabilities may extend beyond supporting internal operations and evolve into independent sources of revenue. Proprietary panel technology could be licensed to external developers or governments seeking to adopt similar infrastructure. The software and control systems that manage energy output and roadway functionality could be deployed as a service offering. Additionally, Colas's accumulated expertise in integrating renewable energy into transportation systems may support consulting or project advisory services. By structuring its operations in this way, the company can both protect its core innovations and create opportunities to scale its capabilities across a broader market.

Technology Commercialization/Collaboration Strategy

Capabilities sought

Given the multidisciplinary nature of solar roadway systems, we would seek several key capabilities through partnerships to complement internal strengths in infrastructure engineering and system integration.

First, photovoltaic technology expertise would be sourced from established solar manufacturers. These partners provide high-efficiency solar cells and benefit from economies of scale, reducing costs and improving performance compared to in-house development .

Second, we would collaborate with electronics and embedded systems providers for components such as sensors, LED lighting, and control hardware. These systems are critical for enabling smart infrastructure features like traffic signaling, condition monitoring, and energy management.

Third, utility companies and grid integration specialists are essential partners. Their capabilities in grid connection, energy distribution, and regulatory compliance ensure that generated electricity can be safely and effectively integrated into existing energy systems.

Additionally, we would seek construction and installation partners, particularly regional contractors with local expertise. While core panel design remains internal, these collaborators enable scalable deployment and efficient installation across different geographies .

We would also partner with government agencies and regulatory bodies to navigate permitting, safety standards, and public infrastructure requirements. Their involvement is critical for pilot programs and large-scale adoption.

Finally, collaboration with software and data analytics firms would support development of control systems, remote monitoring, and optimization platforms. These capabilities enable the transition from a physical product to a smart, service-based infrastructure offering.

Stakeholder	Role
Established photovoltaic manufacturers	Supply high-efficiency solar cells; reduce cost through scale; improve energy performance

Advanced materials firms	Provide durable glass, coatings, and composite materials capable of withstanding traffic and environmental conditions
Industrial engineering / manufacturing partners	Optimize production processes, modular assembly, and system reliability for scalable manufacturing
Utility companies and grid operators	Enable grid integration, manage energy distribution, and ensure compliance with energy regulations
Public sector entities / pilot cities	Support permitting, testing, and deployment through pilot programs and public infrastructure integration

Figure 23: Key Stakeholder/Role Table for Solar Roadways

Prospective collaborators

One clear group of collaborators would be established photovoltaic manufacturers such as First Solar or SunPower. These firms bring decades of experience in improving solar cell efficiency, durability, and cost through large-scale production. Their involvement would ensure that roadway panels incorporate proven solar technologies rather than experimental designs, which is especially important given the harsh mechanical and environmental stresses roads must withstand. Industry research consistently shows that manufacturing scale and incremental efficiency improvements are key drivers of cost reduction in solar deployment, making these partners essential for economic viability (International Energy Agency, 2020).

A second set of collaborators would include advanced materials and industrial engineering firms, particularly companies with expertise in durable composites and specialty glass. Firms like Corning or Saint-Gobain are relevant because solar roadways require surfaces that are both transparent enough for light transmission and strong enough to handle vehicle loads. Studies on solar roadway prototypes highlight that material limitations, especially related to durability and traction, are one of the primary barriers to commercialization (Lv, Zha, Hu, Lei, & Niu, 2025). Partnering with companies that already specialize in high-performance materials would significantly reduce technical risk.

Utility companies and grid operators would also be critical collaborators, as service providers and as strategic partners in deployment. Organizations such as the Tennessee Valley Authority or large investor-owned utilities have the infrastructure and regulatory experience needed to integrate distributed generation into the grid. Their involvement ensures that energy generated from roadways can be reliably transmitted, stored, and monetized. Without this integration, even a technically successful roadway would fail to deliver economic value. The International Energy Agency emphasizes that grid integration is one of the most important challenges for emerging distributed energy technologies, particularly those with intermittent output.

Finally, collaboration with public sector entities and pilot cities would be essential for early adoption. Municipal governments, state transportation departments, and federal agencies provide access to real-world testing environments and funding mechanisms. Early pilot programs are especially important for technologies like solar roadways that require public trust and regulatory approval. Research on infrastructure innovation shows that government-backed pilot projects often serve as the bridge between demonstration and widespread adoption, particularly in energy and transportation systems (International Energy Agency, 2020)

Collaborator assessment

Stakeholder	Dependence	Interests	Posture	Approach
Established photovoltaic manufacturers	H	Commercialize proven solar cells, expand into new markets, and lower costs through scale.	N / SP	Persuade
Advanced materials firms	H	Supply durable transparent glass, coatings, and composites that can survive traffic and weather.	P	Collaborate
Industrial engineering / manufacturing partners	M	Improve manufacturability, modular assembly, reliability, and cost-down processes.	P	Collaborate
Utility companies and grid operators	H	Integrate distributed generation, maintain grid reliability, and create viable revenue pathways.	SP	Persuade + Collaborate
Public sector entities / pilot cities	H	Test infrastructure innovation, meet sustainability goals, access funding, and build public trust.	P	Collaborate

Figure 24: High-level SIPOC diagram highlighting key components of the operational strategy

Figure 24 identifies the probable areas of interest, level of dependence, posture, and engagement strategy of the primary stakeholders in relation to solar roadways. Well-known makers of photovoltaics like First Solar or SunPower would mainly be interested in applying existing solar technology into a new commercial market without losing efficiency, reliability, and production volume. The dependency of the host firm on these stakeholders would be relatively high since bankable know-how regarding photovoltaics and significant production capabilities would be necessary for the transition from the prototype phase of solar roads to a product. A neutral to positive posture could be expected from them as well since the idea of solar roads may be seen as a possible new field for their products although they might also face some risks by placing them on roads instead of traditional locations.

Advanced materials firms such as Corning or Saint-Gobain would be interested in applying specialty glass, coatings, and composite technologies to a high-visibility infrastructure application. Dependence on these firms would also be high because

transparent, load-bearing, skid-resistant, and weather-resistant surfaces are one of the main technical bottlenecks in solar roadway design. Their posture would likely be positive, since the challenge aligns directly with their materials-science capabilities and offers an opportunity to open a premium new market. A collaborative approach would therefore be appropriate, especially through joint prototyping, durability testing, and iterative refinement of panel surfaces.

Industrial engineering and manufacturing partners would mainly be interested in scalable production methods, modular assembly, and process optimization. Dependence on them would be medium because a pilot concept could be developed without them, but commercialization would remain difficult without expert support in manufacturability, quality control, and cost reduction. Their posture would likely be positive because solar roadways offer a new application for automation and industrial process expertise. A collaborative approach focused on design-for-manufacture and cost-down strategies would be most suitable.

Utility companies and grid operators would be interested in safe grid integration, reliable distributed generation, and revenue models that fit within existing regulatory structures. The host organization would have a high dependence on these partners because electricity generated by roadway panels only becomes valuable when it can be transmitted, balanced, stored, and monetized through the grid. Their posture would likely be somewhat positive rather than fully positive: utilities generally support distributed energy, but they are cautious about intermittency, interconnection complexity, and uncertain returns from unproven infrastructure. Accordingly, the best approach would combine persuasion and collaboration through pilot interconnection agreements, early operational data sharing, and transparent business-case analysis.

Finally, public sector entities and pilot cities would be interested in infrastructure modernization, sustainability targets, economic development, and public visibility. Dependence on them would be high because they provide access to real-world test sites, permits, funding mechanisms, and the public legitimacy needed for early deployment. Their posture would likely be positive if solar roadways are framed as controlled pilots with clear public benefits, although they will remain sensitive to safety, cost, and political risk. A collaborative approach is therefore most appropriate, built around grant-backed demonstrations, transparent safety validation, and public-facing pilot programs.

Collaboration form

What collaboration strategy will you seek with each collaborator?

The collaboration strategy for solar roadway systems will vary by partner type, reflecting differences in capabilities, risk exposure, and strategic alignment. Rather than relying on a single model, a combination of collaboration forms should be used to balance control, flexibility, and speed of development.

Photovoltaic manufacturers (e.g., First Solar, SunPower) would primarily be engaged through strategic supplier agreements and co-development partnerships. In early stages, co-development agreements would allow customization of solar cells for roadway-specific conditions such as load-bearing requirements and partial shading. As the

technology matures, the relationship would transition toward long-term supply contracts to leverage economies of scale while maintaining cost predictability.

Advanced materials firms (e.g., Corning, Saint-Gobain) would be best engaged through joint development agreements (JDAs). Given that materials innovation is a core technical bottleneck, close collaboration is required for iterative prototyping, testing, and refinement of transparent, durable surfaces. These partnerships may include shared intellectual property arrangements, particularly where novel coatings or composite structures are developed specifically for roadway applications.

Industrial engineering and manufacturing partners would be engaged through a mix of contract manufacturing and process co-design partnerships. Early collaboration would focus on design-for-manufacturing (DFM), ensuring that panel systems can be produced at scale with high reliability. Over time, this could evolve into contract manufacturing agreements or even dedicated production lines, depending on volume and cost targets.

Utility companies and grid operators would require a combination of pilot project agreements, power purchase agreements (PPAs), and interconnection partnerships. Initial collaboration would center on pilot deployments to validate technical feasibility and grid compatibility. As performance is demonstrated, PPAs would provide a revenue model for electricity generated, while formal interconnection agreements ensure regulatory compliance and operational integration.

Public sector entities and pilot cities would be engaged through public-private partnerships (PPPs), grant-funded pilot programs, and demonstration projects. These collaborations are essential for early-stage deployment, as they provide access to infrastructure, funding support, and regulatory pathways. Structuring these as PPPs allows risk-sharing while aligning incentives around public benefits such as sustainability and infrastructure innovation.

Software and data analytics partners would be engaged through technology partnerships or platform integration agreements. In some cases, this may take the form of outsourcing specific components (e.g., monitoring dashboards), while in others it could involve deeper integration or joint ventures to develop proprietary smart infrastructure platforms. Maintaining flexibility here is important, as software capabilities evolve rapidly and may become a source of competitive differentiation.

Overall, the collaboration strategy emphasizes early-stage co-development and risk-sharing, followed by scaling through standardized supplier and commercial agreements. This staged approach enables the venture to access critical external expertise while gradually building internal capabilities and maintaining strategic control over the system architecture.

Project Valuation & Financing

Profit Model

After considering the implementation of solar roadway technology, the most appropriate profit model is a modified traditional profit model with recurring revenue components. In this model, Solar Roadways generates revenue primarily through the sale and installation of modular solar panels to governments, municipalities, and private developers. Similar to traditional infrastructure projects, customers pay for the value of the road system upfront, which includes durability, energy generation, and smart features such as embedded sensors and lighting. This model works best if the lifecycle cost of solar roadways becomes competitive with conventional asphalt, since this creates a more level playing field against existing solutions.

However, achieving price parity is challenging due to the high research and development costs, as well as the significant expenses associated with manufacturing and installation. Traditional road construction remains relatively inexpensive, which creates a price ceiling that could limit adoption. Despite this, the initial target market reduces this concern. Early adopters such as government agencies and sustainability-focused municipalities are often less price sensitive and more willing to invest in innovative infrastructure that provides long-term environmental and energy benefits. These customers are more likely to evaluate the project based on total lifecycle value rather than upfront cost alone, making the traditional model more viable in early stages.

Additionally, this profit model can be adjusted to better align with how large infrastructure projects are typically financed while also incorporating ongoing revenue streams. Rather than requiring full upfront payment, Solar Roadways can adopt a phased payment structure tied to project milestones such as manufacturing, delivery, and installation. Beyond the initial sale, the company can generate recurring revenue through maintenance contracts, system monitoring services, and software updates for traffic management and energy optimization. These service fees provide a steady income stream over time while ensuring the long-term performance of the infrastructure. By combining upfront installation revenue with ongoing maintenance and service agreements, Solar Roadways can improve scalability, reduce financial risk, and more fully capture the economic value of its technology.

Pro forma financial statement

The following pro forma financial statement outlines the key cost inputs and macroeconomic assumptions for Solar Roadways under Sunstrider Pavement. Because this product integrates photovoltaic generation, structural materials, and embedded infrastructure systems, the estimates reflect a hybrid between construction and energy technology industries. While these values are grounded in industry benchmarks and strategic positioning, they should be validated through pilot deployments and supplier data before large-scale commercialization.

Purchased Inputs: \$140 per unit

Purchased inputs represent the externally sourced materials required to manufacture each solar roadway panel. These include photovoltaic cells, tempered structural glass, embedded lighting components, heating elements, and control electronics. Although these components are complex, they benefit from existing supplier ecosystems in both the solar and materials industries. A value of \$140 per unit reflects early-stage procurement at moderate scale, where some cost efficiencies are achieved through partnerships, but full economies of scale have not yet been realized.

Value-added: \$180 per unit

Value-added costs include internal labor, panel assembly, system integration, testing, and preparation for installation. Because solar roadway panels must meet both structural and electrical performance requirements, these costs are higher than those of traditional paving materials. However, the use of modular panel design and partnerships with construction contractors reduces labor intensity compared to fully custom infrastructure builds. A value of \$180 per unit reflects this balance between complexity and early operational efficiency.

Purchased Input Cost Inflator in Outyears: 2.0%

The purchased input cost inflator represents the expected annual increase in material costs. A 2.0% rate is appropriate because it reflects moderate long-term cost growth in materials and electronics without overstating inflationary pressure. This assumption also accounts for the likelihood that improvements in sourcing and supplier competition will partially offset broader cost increases over time.

Experience Curve Effect: 15%

The experience curve factor reflects the expected reduction in unit cost as cumulative production increases. Solar roadways benefit from both photovoltaic manufacturing learning curves and incremental efficiencies in modular construction processes. A 15% experience curve represents a blended estimate that captures moderate cost reductions as production scales, without assuming overly aggressive efficiency gains.

Capital Investment: \$2.5M (2025), \$4.0M (2026), \$1.5M (2027), \$1.0M (2028)

Capital investment includes expenditures on manufacturing setup, testing infrastructure, installation equipment, and pilot deployment capabilities. Investment peaks around the launch year to support production readiness and initial deployments, then declines as the company transitions to incremental scaling. This staged investment strategy aligns with a pilot-first approach, focusing on controlled environments such as campuses and pedestrian pathways before broader expansion.

Research & Development: \$0.5M (2024), \$2.0M (2025), \$3.0M (2026), \$2.5M (2027), \$2.0M (2028)

Research and development is critical for ensuring durability, safety, and system integration. Solar roadway panels must withstand mechanical loads, environmental stress, and long-term wear while maintaining electrical performance. R&D spending peaks near

launch as prototypes are refined into deployable products and remains significant post-launch to support ongoing improvements in reliability and manufacturability.

Startup Operating Expenses: \$0.3M (2024), \$0.75M (2025), \$1.25M (2026)

Startup operating expenses include administrative costs, permitting, legal support, early business development, and pilot project coordination. These expenses increase leading up to launch as the company prepares for commercialization and establishes relationships with customers and contractors. Although smaller than capital and R&D investments, these costs are necessary to support initial operations.

Post-Launch Operating Expense Factor: 18%

The post-launch operating expense factor reflects ongoing costs that scale with the business, including maintenance support, monitoring systems, customer service, and warranty obligations. Solar roadways are more service-intensive than traditional infrastructure due to their embedded electronics and performance requirements. An 18% factor captures this elevated operational burden while remaining within a reasonable range for infrastructure-based technologies.

Initial Sales Price: \$4,500 per unit

The initial sales price reflects the bundled value provided by solar roadway panels. Unlike traditional pavement, these panels integrate energy generation, embedded lighting, heating for snow and ice mitigation, and potential smart infrastructure capabilities. This multifunctional value allows for premium pricing, particularly in early target markets such as campuses, sidewalks, and controlled-access environments where sustainability and innovation are prioritized. A price of \$4,500 per unit balances competitiveness with the need to recover high initial costs.

Sales Price Inflator in Outyears: 3.0%

The sales price inflator represents expected annual increases in price due to inflation and value improvements. A 3.0% rate aligns with long-term macroeconomic expectations and reflects the ability to maintain pricing power as the technology matures and demonstrates value.

Service Life Assumption: 12 years

Solar roadway panels are expected to have a service life of approximately 12 years. While photovoltaic components can last longer, the structural demands of roadway applications introduce additional wear and degradation factors. This assumption reflects a conservative estimate that accounts for mechanical stress, environmental exposure, and the need for periodic replacement.

General Inflation Rate: 3.30%

The general inflation rate is assumed to be 3.30%, reflecting a stable long-term macroeconomic baseline. This avoids overemphasizing short-term inflation volatility while providing a realistic foundation for cost and revenue projections.

Risk-Free Rate of Return: 4.0%

The risk-free rate is based on current U.S. Treasury yields and represents the baseline return for a riskless investment. A value of 4.0% is appropriate for medium-term financial modeling and aligns with prevailing market conditions.

Company Cost of Capital: 8.5%

The company cost of capital is benchmarked against comparable industries such as engineering and construction, electrical equipment, and renewable energy. These sectors typically exhibit cost-of-equity values in the high single digits, making 8.5% a reasonable midpoint estimate. This reflects both the infrastructure and technology components of the business.

Project Risk Premium: 5.5%

The project risk premium accounts for the additional uncertainty associated with commercializing solar roadway technology. Risks include technical performance, adoption rates, regulatory challenges, and long-term maintenance costs. A 5.5% premium reflects the early-stage nature of the project while remaining within a defensible range.

Project Cost of Capital (Discount Rate): ~21.0%

The resulting project cost of capital is approximately 21.0%, reflecting the combined effects of baseline financing costs, macroeconomic conditions, and project-specific risk. While higher than that of traditional infrastructure projects, this rate is appropriate given the novel and capital-intensive nature of solar roadway deployment.

Discussion of Assumptions and Model Limitations

Several assumptions in this model are subject to uncertainty due to the early-stage nature of the technology. The most critical areas for further validation include component costs, installation labor requirements, long-term durability, and maintenance frequency. Access to supplier quotes, pilot project data, and engineering test results would significantly improve the accuracy of these estimates.

Additionally, the model could be enhanced by separating manufacturing, installation, and maintenance costs rather than aggregating them into broad categories. Incorporating a dedicated maintenance cost structure and segmenting the market by application (e.g., sidewalks, campuses, airports) would also improve realism. Finally, the discount rate formulation may overstate risk by combining multiple factors additively, and could be refined in future iterations.

Results

Based on the financial model, the solar roadway business shows strong long-term demand potential but weak near-term financial performance. Unit sales increase steadily over time; however, early years are dominated by high R&D and capital investment costs, resulting in negative gross margins and consistently negative cash flows. The project will yield a negative NPV and does not achieve payback within the 10-year period. The discounted cash flow analysis indicates that cumulative returns remain below

breakeven, suggesting that a longer time horizon (beyond 10-15 years) is required for true, reliable viability.

The core issue is the timing difference between large upfront costs and delayed growth of revenue. Capital expenses and production costs are concentrated in the early years, while meaningful revenue only emerges later as adoption increases. This creates a capital-intensive profile that is hard to justify under the standard investment criteria without external support, such as sponsors or partnerships.

Improving the financial performance primarily depends on reducing costs. A reduction of approximately 25% in manufacturing and installation costs would improve gross margins, reduce cumulative losses, and shorten the payback period. Additional improvements could come from increasing revenue per unit and accelerating early adoption. Overall, the business is not viable within a 10-year horizon under current assumptions but may become viable with cost reductions and a longer investment timeframe.

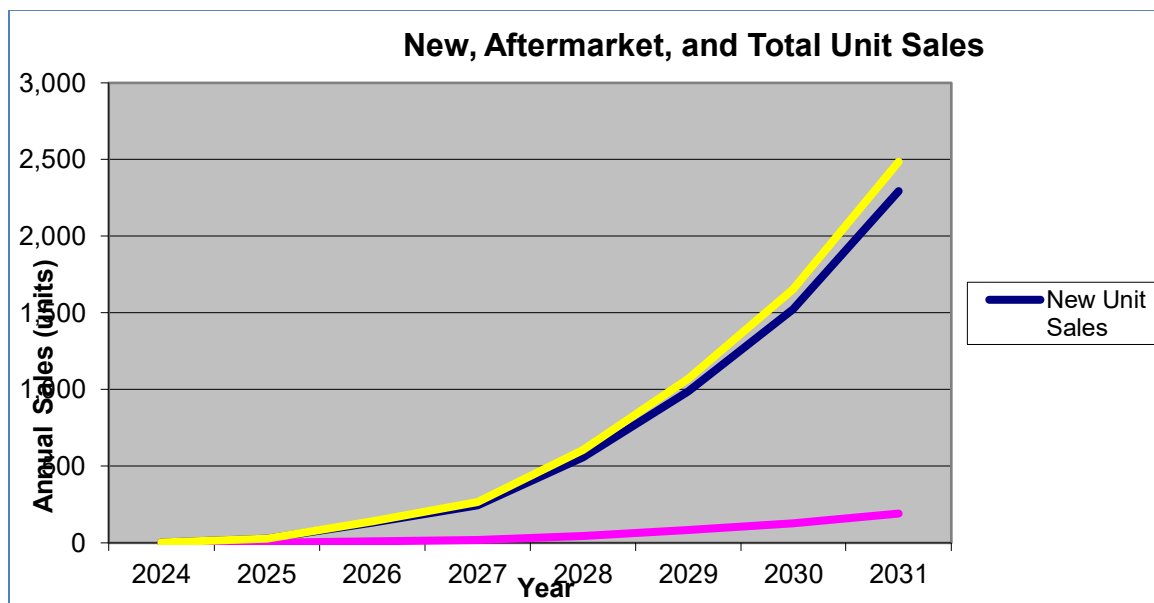


Figure 25: New, Aftermarket, and Total Units Sales

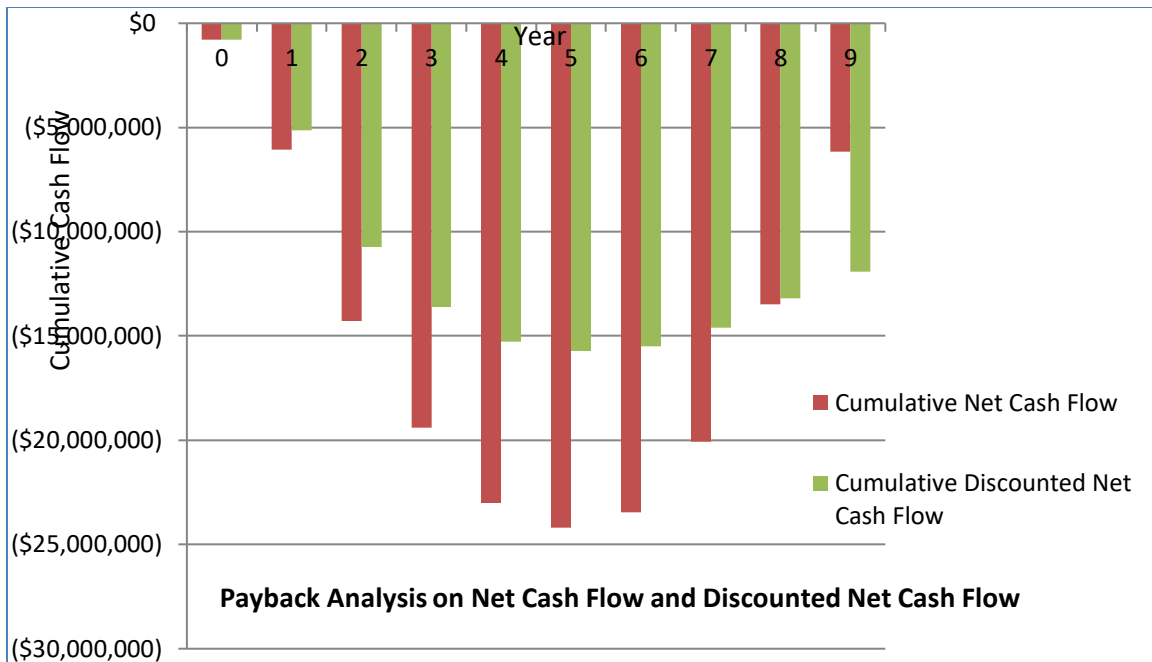


Figure 26: Payback Analysis on Net Cash Flow and Discounted Net Cash Flow

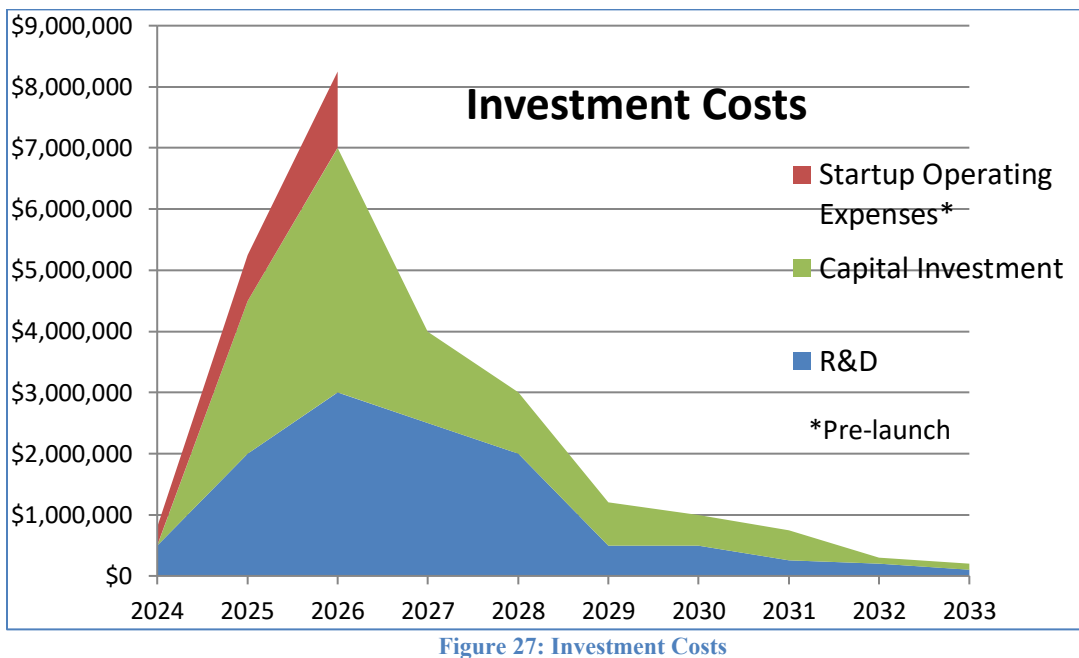


Figure 27: Investment Costs

Capital Requirements and Sources of Capital

Solar roadways are best positioned as a “Question Mark” on the BCG Growth-Share Matrix. The broader renewable energy and smart infrastructure market is growing rapidly, but solar roadways currently have very low market share compared to traditional solar solutions. The technology is still developing in terms of durability and efficiency,

which means it requires significant upfront investment before it can scale. Based on pilot efforts like those from Colas and Solar Roadways, an estimated \$250 million to \$400 million over roughly 6 to 8 years would be required to reach a self-sustaining stage, covering R&D, pilot deployments, manufacturing, and early operations. Figure 12 in Section 4.4: Projected market share shows more information about the projected market share of solar roadways, illustrating a gradual increase in adoption as the incumbent's dominance declines and new entrants, including our company, capture a growing portion of the market over time.

Internal funding alone is unlikely to be sufficient unless backed by a large infrastructure firm, since capital would compete with more proven investments like traditional solar and transportation projects. As a result, external funding is critical. Government grants and public-private partnerships are especially attractive given the alignment with clean energy and infrastructure goals.

Funding should be raised in stages to match technological validation. An initial \$25–50 million round would support testing and small pilot installations to prove reliability. A second \$100–150 million round would expand manufacturing and real-world deployments with partners. A final \$150–200 million round would scale production and move toward profitability. This phased approach aligns with investor expectations and allows the company to demonstrate progress before securing larger commitments.

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Appendices

Contact Log

Who contacted	Contact	Date	Success?	Referrals	Follow-up Actions
Juliette	Scott & Julie Brusaw	4/4	Yes	...	Wrote a follow-up on 4/11
Pruett	Thomas Rush	4/5	Yes	...	Wrote a follow-up on 4/12
Sophie	Dr. Caldwell	4/5	Yes	...	

Interview Summaries

Interview with Dr. Caldwell Writeup:

We spoke with Dr. Caldwell to better understand the technical feasibility of solar roadways, particularly from a physics and materials perspective. While he noted that solar cell design is outside his primary area of expertise, his background in infrared

spectroscopy and optical systems provided useful insight into how light interacts with surfaces and how that affects performance.

When asked how efficient solar panels embedded in horizontal road surfaces would be compared to traditional tilted installations, he explained that the key driver is not necessarily a loss in panel efficiency, but rather a reduction in solar irradiance due to the angle of incidence. As the sun moves throughout the day, the amount of energy reaching a flat surface changes significantly, which can be calculated as a function of angle. He emphasized that without modeling this variation, it is difficult to estimate total efficiency loss, though it is clear that horizontal systems will receive less consistent energy input than optimally angled panels. In response to questions about energy loss from dirt and surface conditions, he noted that while there has been research into coatings that reduce dust and debris accumulation, he was not familiar with the specific effectiveness of these solutions.

On the structural side, when asked about long-term durability concerns such as cracking, impact resistance, and fatigue under traffic, he stressed that durability is highly dependent on the specific material system and structural design. Since solar roadways integrate multiple components such as glass, electronics, and load-bearing layers, performance cannot be generalized across designs. He explained that predicting durability is nearly impossible without first defining the exact materials and configuration, and that real-world testing would be essential. More broadly, when discussing overall feasibility, he emphasized that many of these questions cannot be definitively answered at this stage, and that proving feasibility through prototyping and testing must come before drawing conclusions about large-scale implementation.

Interview with Scott & Julie Brusaw Writeup:

We spoke with the Solar Roadways team to better understand the feasibility of their technology and the challenges associated with scaling the system. Their responses provided a realistic view of both the potential and the limitations of solar roadway infrastructure.

When discussing durability and structural performance, they emphasized that this has been one of the most significant challenges in real-world testing. Unlike traditional solar panels, roadway panels must withstand constant mechanical stress from vehicle loads, temperature fluctuations, and environmental exposure. They explained that designing a surface that is strong enough to support traffic while also remaining transparent for solar energy capture is a complex engineering problem. In addition, they noted that maintaining sufficient traction and safety for vehicles has required extensive testing and iteration. These combined factors have made durability one of the primary constraints on widespread implementation.

In terms of energy generation, they acknowledged that solar roadway panels currently underperform compared to traditional solar installations. They explained that conventional panels are typically installed at optimized angles and in unobstructed environments, whereas roadways are flat and frequently subject to shading from vehicles and surrounding structures. As a result, the overall energy output is lower and less consistent. However, they highlighted that the concept is not solely about maximizing efficiency, but rather about utilizing existing infrastructure to generate energy in spaces that would otherwise remain unused.

When asked about large-scale adoption, they identified cost and infrastructure integration as the biggest barriers. Replacing existing road systems would require significant upfront investment, and long-term maintenance costs remain uncertain. They also pointed out that adoption would likely depend heavily on government support and public-private partnerships. While they remain optimistic about the long-term potential of the technology, they described widespread deployment within the next five to ten years as unlikely. Instead, they see near-term progress occurring through smaller-scale applications such as sidewalks, parking lots, and bike paths.

Overall, their responses emphasized that while solar roadways present an innovative approach to energy generation, there are still substantial technical and economic challenges that must be addressed before the technology can be implemented at scale.

Interview with Thomas Rush Writeup:

We spoke with Thomas Rush, CEO of Electrovia, to better understand the technical and regulatory feasibility of integrating energy-generating infrastructure into roadways. His experience working at the intersection of energy systems and construction provided a grounded perspective on what it would take to scale solar roadway technology. Overall, his responses highlighted that while the concept is innovative, the practical barriers are significant and often underestimated.

When discussing cost and manufacturing, he explained that solar roadway panels would be substantially more expensive than traditional asphalt or concrete, likely several times higher on a per square foot basis. He noted that the most expensive components are the protective materials required to make the panels durable enough for traffic and the embedded electronics for energy conversion. While he acknowledged that costs could decrease with large-scale production, he emphasized that they would still remain far above conventional road construction due to the complexity of the system.

Regarding regulatory and infrastructure integration, he emphasized that deployment would require navigating a complex mix of federal, state, and local approvals, particularly from departments of transportation and energy regulators. He noted that safety certification alone would be a major hurdle, as any roadway technology must meet strict standards for durability and reliability. In addition, connecting distributed roadway panels to the electrical grid would introduce logistical challenges, including managing decentralized energy inputs and ensuring compatibility with existing infrastructure. While

he believes transportation agencies may be open to pilot programs, especially in controlled environments, he suggested that widespread adoption would likely be slow and dependent on strong public-private collaboration.

Overall, his insights suggest that while solar roadway systems are conceptually appealing, their economic viability and technical feasibility remain uncertain. He sees more immediate potential in adjacent innovations, such as embedded roadway technologies for charging or sensing, rather than large-scale energy generation through solar panels.